

China Walls

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Abstract

We evaluate the enforcement of China Walls—information barriers within conglomerates. We focus on July 2018, when the SEC tightened enforcement around broker-dealer subsidiaries, and on the 25 million Israeli Shekel trades over 2018–2024. Our difference-in-differences design compares a dealer’s affiliate, connected, and unrelated funds on days when the dealer likely obtained valuable information. Before July 2018, dealers share information with their affiliate and connected funds. After July 2018: the sharing ceases only with the affiliate funds; price discovery slows; trading costs fall for affiliate funds and rise for other funds. Tightly enforced walls eliminated 99.0% of affiliate funds’ event-day profits.

JEL classification: G21, G28, G14, G15

Keywords: Banking conglomerates, trading networks, information barriers, information sharing, foreign exchange

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1 Introduction

Banking conglomerates are rife with conflicts of interest. These firms analyze securities while underwriting them, manage investment funds while acting as dealers for the funds of others, and much else. Acting on these conflicts often involves affiliated subsidiaries sharing private information among themselves, which contributed to the last financial crisis (Griffin, 2021). In response, regulators tightened information barriers within banking conglomerates, called *China Walls*.¹

Enforcing China Walls is challenging. Bankers already share information in plausibly deniable ways (Peluso, 2020). The China Walls in particular target information sharing within a single conglomerate, which is unlikely to report or litigate against itself. Many policymakers and legal scholars have been skeptical for these reasons (Avci, Schipani, and Seyhun, 2018), with some calling for banking conglomerates to be broken up as an alternative way to remedy conflicts of interest.² Yet, federal securities law empowers the US Securities and Exchange Commission (SEC) to charge broker-dealers that fail to maintain China Walls, whether or not they profited from violating the walls. The SEC tightened its enforcement of the walls in July 2018, when it fined Mizuho Securities \$1.25 million for failing to maintain China Walls without alleging *any trading* based on private information. It is unknown whether today’s tightened enforcement effectively walls off dealers from their affiliates.

We document that China Walls are effectively enforced in the foreign exchange market, where trading on private information is otherwise widespread.³ Dealers intermediate all foreign exchange trades. Given any dealer, each fund is either affiliated via common control, connected via a trading relationship, or entirely unrelated. The China Walls seek to block

¹We use “China Walls” rather than the more common “Chinese Walls” or “information barriers,” because it is concise and closer to the original reference to the Great Wall of China (Gozzi, 2003).

²Among them, former Federal Reserve Chairman Paul Volcker testified that China Walls cannot “remain impermeable against the pressures to seek maximum profit” (Reuters, 2010).

³Foreign exchange order flows predict short-run currency returns (Evans and Lyons, 2002; Froot and Ramadorai, 2005), particularly those of hedge funds and other informed traders (Menkhoff, Sarno, Schmeling, and Schrimpf, 2016; Ranaldo and Somogyi, 2021; Cespa, Gargano, Riddiough, and Sarno, 2022; Hacıoglu Hoke, Ostry, Rey, Rousset Planat, Stavrakeva, and Tang, 2026).

dealers from sharing private information on client orders with their affiliate funds. We compare the trading activities and profits of the affiliate funds against the funds unrelated to a dealer around the days when the dealer obtains unusually valuable order flow information from exceptionally large trades. We detect violations of China Walls if the affiliate funds make more trades or profit around those days relative to the unrelated funds. The period before the Mizuho settlement, alongside the funds connected to the dealer, lets us falsify our design. We detect information sharing between dealers and their affiliate funds only before the settlement, and between dealers and their connected funds throughout. In sum, we detect information sharing everywhere, except where and when the China Walls become tightly enforced.

Section 2 details the empirical and the regulatory context. The Shekel market is large and liquid, with average daily trades worth USD 13 billion and 83% of them in the USD-Shekel currency pair. The largest dealers in the Shekel and in the broader USD market are identical. Israeli law prevents domestic banks from controlling investment funds, leaving no Israeli affiliate dealer-fund pairs in our setting. For the foreign banking conglomerates that do combine dealers and funds, the relevant China-Wall enforcement comes primarily through the SEC’s authority over US-registered broker-dealers.

Section 3 describes the data. We obtain the 25 million foreign exchange trades involving the Israeli Shekel from 2018 through 2024. Dealers frequently trade with their affiliate funds before July 2018, soon after which they virtually *never* do so. No such discontinuity exists between dealers and their nonaffiliate connected funds.

Section 4 explains our difference-in-differences design, illustrated in **Figure 1**. An event is a dealer-day on which the dealer makes at least one trade in the largest 0.1% of that dealer’s trades by USD value. We compare the daily gross dollar volume, net volume in the opposite direction of the dealer’s event trade, and one-week future profit-and-loss (P&L) of the dealer’s affiliate funds against nonaffiliate funds that never trade with the dealer (unrelated funds) around the event day.

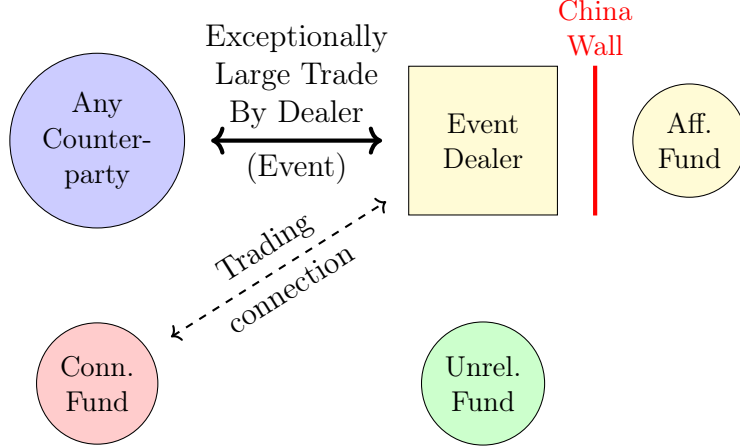


Figure 1: Identifying Information Sharing from Dealers to Affiliate Funds

This design addresses two challenges. First, we can rule out China Wall violations without examining every possible setting. Under the assumption that larger trades contain more valuable information, each dealer has the strongest incentive to violate China Walls on the days when it intermediates its largest trades. Detecting no information sharing around these events plausibly rules out sharing in other times, when dealers have a weaker incentive to do so. Our assumption is supported by standard theory (Kyle, 1985; Easley and O’Hara, 1987), evidence in other over-the-counter markets (Kumar, Mullally, Ray, and Tang, 2020; Pinter, Wang, and Zou, 2024), and a set of identification tests in our own market.

The second challenge is to detect information sharing where it exists, without falsely detecting it where it does not exist. Using the unrelated funds as controls preserves any increase in volumes or profit due to information sharing, under the plausible assumption that dealers do not share private information with funds that are neither clients nor affiliates. At the same time, the unrelated-fund control removes the spurious increases from inventory management. The event trade leaves the event dealer or its counterparty with inventory to lay off, impacting exchange rates and liquidity. Since these impacts would affect the affiliate and the unrelated funds alike, our control partials them out. Likewise, the control partials out macroeconomic news, fluctuations in currency demand, and other shocks common to all funds. To eliminate the remaining spurious variation from the event dealer’s trades around

its event day, we exclude any fund–event pair whose fund traded with the event dealer on the event day or within the following five days.

The connected funds let us falsify whether our design reliably detects information sharing. Unlike affiliate funds, connected funds are not walled off from their dealers, which are well known to share information with their clients (Barbon, Di Maggio, Franzoni, and Landier, 2019; Boyarchenko, Lucca, and Veldkamp, 2021). Dealers in foreign exchange markets may legally share aggregated and anonymized order flow information, and our conversations with them suggest that they regularly share such “market color” information with their clients. We thus expect a reliable design to consistently detect information sharing between connected dealers and funds.

Section 5 implements our design as stacked difference-in-differences specifications with never-treated controls (Cengiz, Dube, Lindner, and Zipperer, 2019) in the 2019–2024 period after the Mizuho settlement. The funds affiliated with the event dealer are statistically and economically indistinguishable from the unrelated funds, with the event-day change in gross volumes differing by 0.017 standard deviation (clustered std. error: 0.011 sd). In contrast, the funds connected to the event dealer increase their gross volumes by a significant 1.95 sd (std. error: 0.002 sd) on the event day relative to the unrelated funds. Neither group exhibits pre-trends in the week prior to the event day. All results remain when we replace gross volumes with net volumes in the opposite direction of the dealer’s event trade or with one-week future P&L. Across fund, event-trade, and price characteristics, connected-fund responses are consistent with information sharing rather than price or inventory impacts of the event trades.

We reject two threats to identification. First, informed traders may split orders to disguise private information (Seppi, 1990). Second, regulators might focus attention on large trades (DeMarzo, Fishman, and Hagerty, 1998), pushing dealers to hide violations among smaller trades. We find that the event trades predict exchange-rate changes over subsequent trading days, whereas smaller trade-size bins are not predictive, and that connected funds respond

only around the event trades. In the sample after the Mizuho settlement, we detect no affiliate-fund response around any trade-size bin. These findings are inconsistent with order splitting or violations being hidden in smaller trades, and consistent with the event trades revealing valuable private information to the dealers.

Section 6 examines the quarters around the Mizuho settlement in July 2018. The event-day increase in affiliate funds' gross volume exceeds that of unrelated funds by 0.42 sd (std. error: 0.030 sd) in 2018Q1, and by 0.31 sd (std. error: 0.034 sd) in 2018Q2. The increase in their P&L exceeds that of unrelated funds by 0.18 sd and 0.22 sd in the same quarters. The affiliate-fund response shrinks in 2018Q3 and becomes economically and statistically insignificant by 2018Q4. The affiliate funds immediately cease responding to the events by US-domiciled dealers, which are more exposed to the SEC, and only cease responding to the non-US dealers' events with a one-quarter delay. The connected-fund responses remain large and stable throughout 2018 and into 2019. Together, these results point to tightened enforcement around the Mizuho settlement as the driver of our findings. A back-of-the-envelope calculation shows that the tightly enforced China Walls eliminated \$8.8 million of the affiliate funds' \$8.9 million event-day profits (99.0%).

Section 7 examines how the tightened enforcement affected prices and trading costs. We find suggestive evidence that price discovery slowed, with the event trades predicting exchange-rate movements several days into the future after the Mizuho settlement, whereas beforehand the predicted movement occurred entirely on the event day. Effective spreads narrowed by 4.5 bps for conglomerate-affiliated funds, consistent with the China Walls reducing their informational advantage. Spreads widened by 2.9 bps for the other funds, as dealers chase order flow less aggressively once they can no longer monetize client information through their affiliate funds.

Our findings do not imply that it is easy to regulate information sharing. We identify information sharing in the aggregate, across all the dealers' event trades in periods of one quarter or longer. Any suspected sharing within a single dealer-fund pair in isolated incidents

remains plausibly deniable. Indeed our finding that affiliate dealers and funds routinely shared information before the Mizuho settlement supports the SEC’s focus on prosecuting inadequate safeguards rather than information sharing itself.

Prior work. Existing evidence on China Walls identifies extensive breaches using samples that predate 2018.⁴ We replicate this result in early 2018, then show that it disappears under the tightened enforcement.⁵ As importantly, we contribute a novel identification strategy that uses unrelated funds as controls to isolate the effects of information sharing. We validate our strategy in conditions where information sharing has been documented in the literature. Applying this design to a large and granular dataset yields precise estimates and robust evidence that today’s China Walls effectively preempt information sharing.

Broader contributions. We belong to the literature on information diffusion in financial markets. Dealers extract information from their clients’ order flow (Hortaçsu and Kastl, 2012), leak information to certain clients (McNally, Shkilko, and Smith, 2017; Barbon et al., 2019; Boyarchenko et al., 2021; Chague, Giovannetti, and Herskovic, 2023), and generally act as the conduits through which information diffuses throughout their trading networks (Di Maggio, Franzoni, Kermani, and Somnavilla, 2019; Hagströmer and Menkveld, 2019; Kumar et al., 2020). We identify a stark void in this informational network driven by regulatory intervention, thereby adding China Walls as a promising source of variation in information flows that is especially relevant today, when the financial sector is highly concentrated in banking conglomerates.

Roadmap. Section 2 details the regulatory context and the empirical setting. Section 3 describes the data and provides preliminary evidence. Section 4 develops the empirical

⁴Lehar and Randl (2006), Irvine, Lipson, and Puckett (2007), Seyhun (2008), Massa and Rehman (2008), Chen and Martin (2011), Ivashina and Sun (2011), Li (2018), Li, Mukherjee, and Sen (2021), Kondor and Pintér (2022), and Haselmann, Leuz, and Schreiber (2023) find evidence of China Wall violations in various settings. The latest in-sample year among them is 2017.

⁵Garrett (2024) and Beck, Silva-Buston, and Wagner (2025) also document the effectiveness of recent regulations on banking conglomerates. Garrett (2024) finds that a Dodd-Frank ban on concurrent advising and bond underwriting for the same municipality lowers the financing costs of municipalities. Beck et al. (2025) find that supervisory cooperation agreements between national regulators cause banking conglomerates to shift lending to subsidiaries in countries without such agreements. We examine limits on information sharing, as opposed to limits on concurrent services or geographic coverage of regulation.

design. [Section 5](#) investigates whether the China Walls are effectively enforced in the Israeli Shekel market. [Section 6](#) investigates whether tightened enforcement reshaped information sharing. [Section 7](#) examines how the tightened enforcement affected prices and trading costs. [Section 8](#) concludes.

2 Context

[Section 2.1](#) defines the China Walls and describes their enforcement by the SEC since 1988, including the Mizuho settlement. [Section 2.2](#) presents the foreign exchange market in the Israeli Shekel.

2.1 Regulatory Context

Material private information (MPI) is any information that *(a)* a reasonable investor would find important for her investment decisions and *(b)* is not publicly disclosed.⁶ MPI includes, for example, proprietary analysis and clients’ order flows. Among the MPI, it is illegal for a dealer to share the order information of specific clients. It is generally permissible to share an aggregated and anonymized summary of those orders as market color. For example, it is illegal to share “Hedge Fund A will buy USD 500 million today,” and legal to share the market color statement, “hedge funds are on net buying USD today.”

China Walls are collections of rules and physical barriers that aim to block the flow of MPI between business units within a financial conglomerate. We study the China Walls between dealers and funds in the same conglomerate. Typical China Walls are blunt instruments that isolate walled-off subsidiaries with separate entrances and opaque, soundproof partitions, restrict their access to shared databases and communication software, and monitor and record employee communications. Because of such walls, affiliate dealers and funds would

⁶Material non-public information (MNPI) is more commonly mentioned in law. MNPI excludes analyses purely based on public information, whereas MPI expressly includes such analyses. We focus on MPI, because we are agnostic about the source of private information, whether proprietary analysis or inside information.

be severely impaired in sharing market color, although doing so would be legally allowed. We expect tightly enforced China Walls to block most forms of information sharing between affiliate dealers and funds, including market color.

The SEC is responsible for the China Walls through its authority over the internal organization of registered broker-dealers. This authority covers US-registered broker-dealers, regardless of the asset class they trade. The foreign-exchange dealers' practices are also governed by the FX Global Code, a voluntary code of good practice ([Section B.4](#)), alongside rules that apply to broker-dealers in general, including the China Walls.⁷

Legal precedent. Regulators began enforcing the China Walls with the 1988 Insider Trading and Securities Fraud Enforcement Act (ITSFEA). ITSFEA granted the SEC (under Exchange Act §15(g), numbered 15(f) until the 2010 Dodd-Frank Act) the risk-based enforcement power to penalize conduct that raises the risk of insider trading, without the need to prove that insider trading occurred. Today, the SEC has jurisdiction over all worldwide activities of US-registered broker-dealers, and monitors their China Walls as a part of its risk-based enforcement against insider trading. On July 23, 2018, the SEC signaled tighter enforcement of China Walls in its settlement with Mizuho Securities. The settlement accused Mizuho of failing to maintain and enforce China Walls around a dealer desk. The SEC did not allege any trading based on the information the desk shared, nor a systemic violation involving numerous accounts over several years ([US Securities and Exchange Commission, 2018](#); [Barrack, Moskowitz-Hesse, Richards, and Cox, 2020](#)). Every previous §15(g) case involving a broker-dealer trading desk alleged one or the other. More recently, the SEC likewise did not allege any trading or profit in its case against Virtu Americas, which agreed to a \$2.5 million settlement in December 2025 ([US Securities and Exchange Commission, 2025](#)). [Appendix A](#) details regulatory history, other rules on banking conglomerates, and recent cases. [Appendix B](#) sets the Mizuho settlement apart from the other China Wall cases.

⁷While the Commodity Futures Trading Commission (CFTC) enforces anti-fraud rules involving spot foreign exchange trades, the CFTC does not regulate broker-dealers' internal organization.

2.2 Empirical Setting

The *foreign exchange market* is an over-the-counter market, in which trades occur bilaterally between dealers or a dealer and one of its clients. Client–dealer trades are nonanonymous, and most clients rely exclusively on one or a few relationship dealers. This market operates at high frequency, where news is rapidly incorporated into exchange rates (Menkhoff et al., 2016). We expect private advantage from an MPI to dissipate in one or a few trading days.

Our empirical setting is the *Israeli Shekel* (ILS) market, whose trading data we obtain from the Bank of Israel (BOI). The ILS market structure closely resembles that of other major foreign exchange markets. Some 83% of ILS transactions involve the USD, similar to the USD share across all foreign exchange transactions (Somogyi, 2022). The ILS and the USD markets have the same largest dealers. Israeli law prevents banks from controlling investment funds (Shemesh and Abir, 2024, p. 191), as the US Glass-Steagall Act did until 1999. The Israeli regulators neither require nor monitor China Walls, perhaps because every Israeli dealer is affiliated with a bank, and therefore cannot be affiliated with a fund. The relevant regulator is the SEC, which reaches the non-Israeli dealers in our sample primarily through its authority over their US-registered broker-dealer entities.

Our setting is a conservative laboratory to evaluate the enforcement of China Walls. Banking conglomerates may report any trade at odds with their China Walls to the BOI without incriminating themselves, because the BOI does not enforce China Walls. Moreover, Israeli dealers and funds cannot drive our results, because no Israeli fund is affiliated with a dealer. Otherwise, the ILS market shares the same structure and the same largest dealers as other major foreign exchange markets. We conclude that ours is the foreign-exchange dataset in which one is most likely to detect a violation.

3 Data

Section 3.1 describes the raw data and trading volumes. Section 3.2 determines the affiliations, connections, domiciles, and fund types. Section 3.3 constructs the analytical sample. Section 3.4 defines the dependent variables.

3.1 Raw Data

We obtain the near universe of foreign exchange trades involving the Israeli Shekel from the BOI for the sample period from January 2018 through March 2024.⁸ Each trade specifies the currency pair (ILS and another currency), the exchange rate, quantity, date and time, asset class (spot, forward, swap, or option), the legal names of counterparties, their trade directions, whether each counterparty is a financial firm (i.e., dealers and investment funds), and whether it is domiciled in Israel. We do not use the time stamps, as the BOI does not verify them and 6.6% of trades report exactly 00:00:00, suggesting placeholder values.

Table 1: Unique Financial Firms in the Raw Data

	Conglomerates	Dealers		Funds		
		All	No affiliate fund	All	No affiliate dealer	Hedge fund
US	13	94	0	4824	4307	364
Europe & UK	13	101	0	1380	968	88
Israel	11	18	18	192	192	2
Elsewhere	9	22	0	1379	1193	178
Total	46	235	18	7775	6660	632

Dealers are financial firms on the BOI's list of foreign exchange dealers. *Funds* are all other financial firms. *Conglomerates* are holding companies that each controls two or more financial firms.

⁸All Israeli banks, including the Israeli branches of non-Israeli banking conglomerates, must report all their ILS trades to the BOI. Any other dealer falls under the same reporting requirement if its ILS trades in the preceding 12 months average at least \$15 million per day. For the dealers that fall below this threshold, we only observe their ILS trades with the dealers that must report to the BOI. We observe all ILS trades by 34 of the 46 dealers in our sample. The BOI can enforce its worldwide reporting requirements through Israeli banks, which must verify that their counterparties are in compliance with BOI rules. Any non-Israeli firm trading ILS has a strong incentive to abide by BOI rules, because Israeli banks maintain the largest holdings of ILS. The BOI reporting requirements can be retrieved from <https://www.boi.org.il/en/economic-roles/statistics/reports-to-bank-of-israel/reporting-on-activity-in-the-foreign-currency-derivative/>.

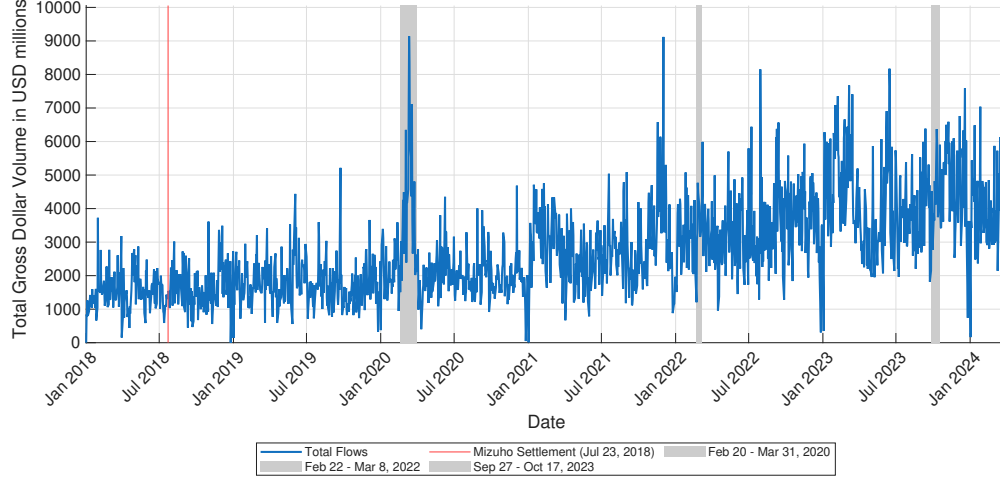
[Table 1](#) describes financial firms in the raw data. *Dealers* are financial firms on the BOI’s list of foreign exchange dealers. *Funds* are all other financial firms. *Conglomerates* are holding companies that each controls two or more financial firms. The plurality of dealers and most funds are in the US with the others mainly in Europe or the UK. There are 46 conglomerates, each of which controls one or more dealers and altogether control all dealers. Most conglomerates control several funds, though most funds are independent. All conglomerates without a fund are Israeli.

[Figure 2](#) plots the total daily gross dollar volume of trades between dealers and funds. Regular falloffs in volume correspond to end-of-year holidays. Nonaffiliated dealer–fund pairs trade USD2.8 billion per day on average ([Figure 2a](#)), and this volume gradually increases throughout the sample period. Affiliate dealers and funds traded regularly before the announcement of the Mizuho settlement on July 23, 2018. They virtually ceased trading soon thereafter, making only four trades worth USD5.51 million from 2019 through 2024. That affiliate dealers and funds regularly traded before July 2018 implies communication between walled-off affiliates. Their ceasing trading with each other is consistent with tightened China Walls. It does not rule out China Wall violations, which we test for below. We can attribute the delay in ceasing of trades to the time needed to interpret the settlement, update dealer workflows, and unwind derivative transactions. The first relevant legal analysis we found is dated July 31st ([Morrison Mahoney LLP, 2018](#)). The first by a national law firm is dated August 7th ([Patterson Belknap Webb & Tyler LLP, 2018](#)).

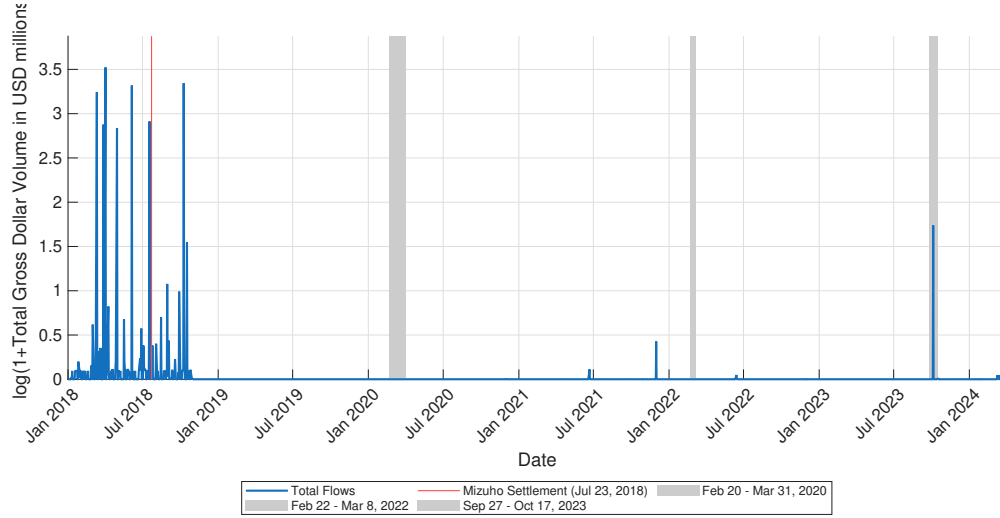
3.2 Affiliations, Connections, Domiciles, and Fund Types

A *connected* dealer-fund pair are unaffiliated and trade at least ten times with each other in our sample. An *unrelated* dealer-fund pair are unaffiliated and never trade with each other in our sample.

We determine affiliations in four steps. First, dealers and funds with obviously indicative names (e.g., “[major bank] Luxembourg S.A.”) are assigned to the corresponding con-



(a) Unaffiliated Dealers and Funds



(b) Affiliate Dealers and Funds

Figure 2: Daily Gross Dollar Volumes Traded Between Dealers and Funds

Figure 2a: The sum of daily gross dollar volume in USD millions across pairs of dealer and fund that are not affiliated with the same banking conglomerate. **Figure 2b:** The sum of daily gross dollar volume across pairs of affiliate dealer and fund, plotted as $\log(1 + \text{volume in USD millions})$. Shaded regions mark the onsets of the Covid pandemic, the Russian Invasion of Ukraine, and the Hamas attack on Israel.

glomerate. Second, we obtain the affiliations of most US-based dealers and funds from the National Information Center database (<https://www.ffiec.gov/npw/>). We assign these affiliations as of the fourth quarter of 2023, because dealers and funds rarely change affiliations, and when they do, they typically change legal names. Third, for the remaining dealer and fund legal names, we query GPT-4o using prompts of the form: “as of [date legal name last appears in the sample], does [legal name] belong to a financial conglomerate or

is it independent? Which holding company, if any, does [legal name] belong to?” Fourth, we manually verify each answer from step three by searching public sources for the legal name paired with “independent” or with the GPT-4o-suggested holding-company name.

We determine each dealer’s and fund’s legal domicile, and classify each fund as a hedge fund or non-hedge fund, using the same query-and-verification procedure. The prompts are: “as of [date legal name last appears in the sample], what is the legal domicile of [legal name]?” and “as of [date legal name last appears in the sample], is [legal name] a hedge fund?”

3.3 Sample Construction

We construct our analytical sample in three steps. First, we drop all option trades, because there are too few observations, and all second legs of swap trades, to avoid double counting.⁹ Second, we consolidate the dealers up to the conglomerate-level by dropping all trades between affiliated dealers and combining the dealers’ trades under conglomerate-level labels. Doing so treats affiliated dealers as a single economic entity, given their ability to split incoming orders and transfer assets and capital among themselves for tax and balance sheet purposes (Bank for International Settlements, 2022). Affiliated funds cannot transfer among themselves, because they manage the capital of their investors. Third, we aggregate the trades of each fund up to the daily frequency. Section 3.4 details the aggregation method for each dependent variable.

Table 2 summarizes each of the three steps. The final sample for analysis comprises 12 million fund-by-day observations. The funds trade more USD and less swaps than the dealers.

⁹A foreign exchange swap combines an initial exchange of currencies with an agreement to reverse that exchange at a predetermined rate on a future date. The first leg is the initial exchange, and the second leg is the reverse exchange. For example, consider a swap today in which the trader buys one USD using ILS at a 3.03 USD/ILS spot rate (first leg; USD/ILS means ILS per USD by convention), and promises to sell one USD for ILS at a 2.86 USD/ILS rate in five days (second leg). This trade would yield a profit if the USD/ILS rate falls below 2.86 in five days. To see this, the trader can immediately sell the USD from the first leg for 3.03 ILS; then buy back one USD just before the second leg using, say, 2.78 ILS; then deliver the USD for 2.86 ILS. Overall profit is $-3.03 + 3.03 - 2.78 + 2.86 = 0.08$ ILS.

Table 2: Each Step in Sample Construction

	Obs.	Unique dealers	Unique funds	Mean value (USD millions)		Share of trades (%)		
				Dealer obs.	Fund obs.	USD	Forward	Swap
Raw data	25,525,488	235	7,775	3.69	1.73	82.8	15.1	14.7
Drop								
options	25,315,204	235	7,775	3.70	1.73	82.8	15.2	14.8
2nd legs	23,372,145	235	7,775	2.45	1.32	82.7	16.5	7.7
Consolidate								
aff. dealers	20,733,476	46	7,775	2.12	1.32	82.6	17.1	7.8
Aggregate to fund-by-day	12,688,800	46	7,775	—	0.32	93.1	21.5	2.8

Mean value is in USD millions and is a simple average across observations. *Share of trades* is the percentage of observations with the corresponding characteristic (i.e., is in the USD/ILS currency pair, a forward trade, or a swap trade). *Raw data* includes all foreign exchange trades reported to the Bank of Israel between January 2018 and March 2024. We *drop* all option trades and all second legs of swap trades (see [Footnote 9](#)). *Consolidate aff. dealers* involves dropping all trades between dealers affiliated to each other and combining the affiliated dealers up to the conglomerate-level. *Fund-by-day* is the analytical sample, in which an observation is the aggregate across all trades and asset classes by a fund on a given trading day in USD terms.

Table 3: Summary Statistics of the Trading Network

	Mean	Std. Dev.	Min	25%	50%	75%	Max
<i>Dealers (N = 46)</i>							
Affiliate funds	24.24	39.88	0.00	0.00	10.50	30.00	182.00
Connected funds	181.89	329.24	0.00	0.00	8.00	178.00	1248.00
Unrelated funds	7251.24	869.94	4747.00	7040.00	7756.50	7775.00	7775.00
<i>Funds (N = 7775)</i>							
Affiliate dealers	0.14	0.35	0.00	0.00	0.00	0.00	1.00
Connected dealers	1.08	1.82	0.00	0.00	0.00	1.00	16.00
Unrelated dealers	42.90	3.16	28.00	41.00	45.00	45.00	46.00

Number of affiliate, connected, and unrelated counterparties per dealer and per fund. Each dealer–fund pair is classified as affiliate (common conglomerate control), connected (trading at least ten times with each other in our sample), unrelated (never trading), or none (trading between 1 and 9 times).

Table 3 describes the network structure in the analytical sample. Each dealer–fund pair is classified as affiliate (common conglomerate control), connected (trading at least ten times with each other in our sample), unrelated (never trading), or none (trading between 1 and 9 times). We classify connections over the full 2018–2024 sample, because dealer-to-client trading relationships are highly persistent (Hendershott, Li, Livdan, and Schürhoff, 2020). Most dealers have at least one affiliate fund and are connected to several funds. The overwhelming majority of funds are not affiliated with any dealer and are connected to at most one dealer. Fifteen of the 46 dealers are unrelated to every fund. These dealers either fall below the BOI reporting threshold, or trade only with nonfinancial firms or with other dealers.

3.4 Events and Dependent Variables

We define *events* in three steps. First, we convert each dealer’s trades to USD using contemporaneous exchange rates. Second, an *event trade* is a trade in the largest 0.1% of that dealer’s trades by USD value. Third, an *event* is a dealer-day on which the dealer makes one or more event trades. We call the dealer the *event dealer* and the day the *event day*. An event takes on the characteristics of its event trade, or of the largest event trade by USD value if the event contains multiple such trades. We set the direction of each swap trade as the direction of the dealer’s economic exposure, which is the opposite of its first-leg direction (Footnote 9).

Figure 3 plots the distribution of events over time. The event frequency is roughly constant. No specific period is likely to drive our results.

Gross dollar volume of fund f on day t aggregates the fund’s daily spot, forward, and swap trades. For each (f, t) -pair, we add together the notional amounts of the spot, forward, swap (first legs only) trades in USD terms. We convert all trade values to USD at the contemporaneous exchange rate.

Net dollar volume in the opposite direction of the dealer’s event trade by fund f on

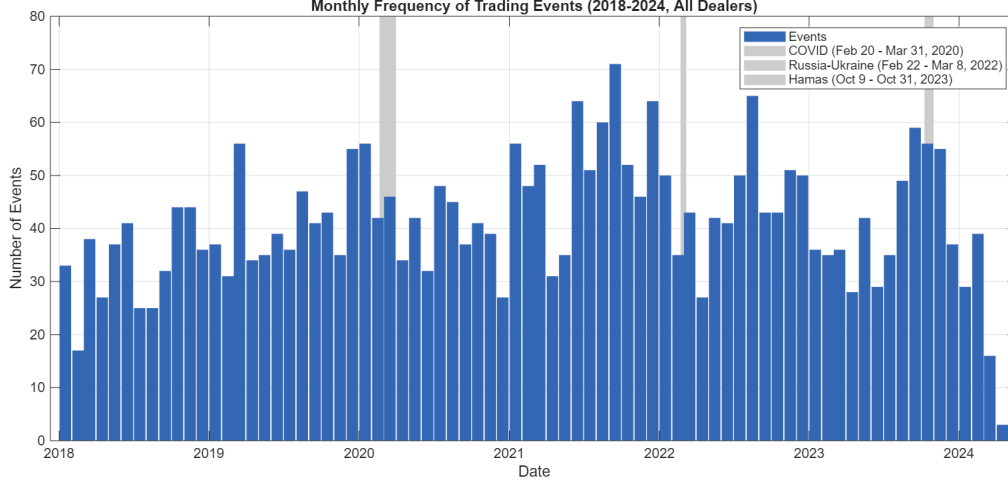


Figure 3: Distribution of Events over Time

day t in event e is computed in two steps. For each $(f, t, e, \text{currency pair})$ -quadruplet, we compute a separate net notional amount (i.e., buys less sales of the non-ILS currency for spot and forward and the opposite for the first legs of swaps; see [Footnote 9](#)) for each non-ILS currency (CCY) in USD terms. We multiply the net notional amount of a given CCY by -1 if the event dealer bought the CCY in the event trade, so that a positive net volume means f traded in the opposite direction of the dealer’s event trade. The net dollar volume is the sum of the net notional amounts across CCY by f on t in e .

One-week future profit-and-loss is the dollar net profit from trades by fund f on day t if those positions were held for five trading days. For each $(f, t, \text{currency pair})$ -triplet, we compute the net notional amount for each CCY in CCY terms. The five-trading day realized return for a given CCY is the proportional change in the exchange rate, $R_t^{sf}(\text{CCY}) := [(CCY/ILS)_{t+5}/(CCY/ILS)_t - 1]$, using the rates at 17:00 New York time on days t and $t+5$ from Bloomberg.¹⁰ We multiply each CCY net notional amount by the corresponding realized return, then convert the resulting amounts into USD using the 17:00 New York time exchange

¹⁰Although the foreign exchange (FX) market is open around the clock, 17:00 New York time is the beginning of the sole interval in a trading day when all major FX markets are closed. It also avoids the various FX fixes throughout the day, around which the USD predictably appreciates then reverses ([Krohn, Mueller, and Whelan, 2024](#)).

rate on $t + 5$ to obtain the net P&L in the CCY by f on t . The one-week future P&L is the sum of the converted net P&L across CCY by f on t .

Table 4: Summary Statistics of the Analytical Sample

	Mean	Std. Dev.	Min	25%	50%	75%	Max
<i>Events (N = 3096)</i>							
Event trade value	253.00	143.56	1.53	199.40	233.61	296.73	1750.00
Has multiple event trades	0.11	0.31	0.00	0.00	0.00	0.00	1.00
Crisis	0.06	0.23	0.00	0.00	0.00	0.00	1.00
USD	0.89	0.31	0.00	1.00	1.00	1.00	1.00
JPY	0.03	0.17	0.00	0.00	0.00	0.00	1.00
EUR	0.04	0.20	0.00	0.00	0.00	0.00	1.00
Spot	0.18	0.38	0.00	0.00	0.00	0.00	1.00
Forward	0.13	0.33	0.00	0.00	0.00	0.00	1.00
Swap	0.70	0.46	0.00	0.00	1.00	1.00	1.00
<i>Fund-by-Day (N = 12,688,800)</i>							
Has trade	0.04	0.21	0.00	0.00	0.00	0.00	1.00
Gross volume	0.36	8.36	0.00	0.00	0.00	0.00	4210.10
Net volume	0.00	2.66	-1201.00	0.00	0.00	0.00	1046.75
1-week future P&L	0.00	0.07	-32.25	0.00	0.00	0.00	35.65

All volumes, P&L, and values are in USD millions. Each event takes on the characteristics of its event trade. If an event includes multiple event trades, the event takes on the characteristics of the largest trade among them by USD value.

Table 4 summarizes the events and the dependent variables in the analytical sample. Event-trade values, volumes, and P&L are in USD millions. Event-trade values vary widely, and event trades are mostly swaps and mostly in the USD/ILS pair. Only a small proportion occur during crisis periods, including the onset of the Covid pandemic (February 1 to March 31, 2020), the Russian invasion of Ukraine (February 24 to March 31, 2022), and the Hamas attack on Israel (October 9, the first trading day after the attack, to October 31, 2023). Some 95% of fund-by-day observations have no trade, because most funds do not trade on a given day. Funds on average neither earn a trading profit nor accumulate a net position in ILS.

4 Design

[Section 4.1](#) describes and explains our empirical design. [Section 4.2](#) provides a test of our central identification assumption.

4.1 Empirical Design

We overcome three challenges to test the hypothesis that the China Walls are effectively enforced. First, it is infeasible to evaluate enforcement in all circumstances. Second, any proxy for bilateral sharing of MPI can be contaminated by common shocks and by the market impacts of the event trade itself. Third, our design may fail to reliably detect bilateral MPI sharing.

Large trades carry valuable MPI. To overcome the first challenge, we seek days on which a dealer receives unusually valuable MPI, under the assumption that information sharing is most likely to occur when gains from sharing are largest. Order flow carries private information in foreign exchange markets ([Evans and Lyons, 2002](#); [Froot and Ramadorai, 2005](#)). A dealer gleans that information from its counterparties' order flow, and plausibly receives its most valuable MPI on the days when it intermediates its largest trades. Standard theory also predicts that an informed trader requests to trade larger quantities when she holds more valuable private information ([Kyle, 1985](#); [Easley and O'Hara, 1987](#)). Empirically, unusually large trades by the same trader are particularly predictive of returns ([Kumar et al., 2020](#); [Pinter et al., 2024](#)). We therefore define an event as a dealer-day on which the dealer makes a trade that is exceptionally large compared to its other trades. [Section 3.4](#) states the precise definition. [Section 4.2](#) develops a joint test of our assumptions that (i) larger trades carry more valuable MPI, and (ii) information sharing is most likely to occur when gains from sharing are largest. [Section 5.3](#) implements the test and concludes that the assumptions hold in our setting.

Isolating information sharing. We consider a dealer d and one of its affiliate funds f

in the days surrounding an exceptionally large trade by d . A proxy for MPI sharing between d and f must isolate information that is (i) material and (ii) shared bilaterally, rather than indirectly through the market. We view the event trade as a source of MPI for d . An informed counterparty requests a trade that is exceptionally large for d , partly revealing the counterparty's private information to d , which may pass this information along to f . Receiving any material information would prompt f to trade in the same direction as the informed counterparty, opposite to d 's side of the event trade. Such trades would earn f positive returns. In contrast, a fund absorbing the event dealer's inventory would trade in the same direction as the dealer did in the event trade. If instead d itself is the informed trader in the event trade, then d sharing its information with f would prompt the fund to trade in the same direction as d . In sum, f trading in the opposite direction of d 's event trade would rule out both f absorbing d 's inventory and d being the informed trader. We look for increases in the daily gross dollar volume, net dollar volume in the opposite direction of the dealer's event trade, and one-week future P&L of f around an event trade by d to proxy for MPI sharing between d and f .

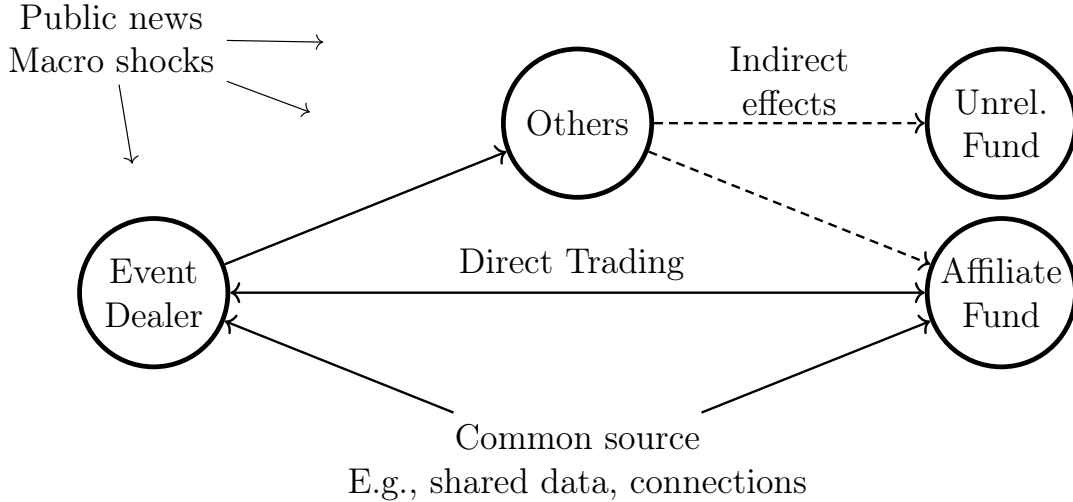


Figure 4: Potential Confounders to Measuring Bilateral Information Sharing

We remove four confounders that could simultaneously induce an exceptionally large trade by dealer d and raise the trade volumes or profit of its affiliate fund f . **Figure 4** illustrates the

confounders. First, any direct trade between d and f would cause a mechanical correlation in their gross trade volumes. We avoid mechanical correlation by excluding any f that traded with d on or in the five trading days after the event day.

A control group addresses the next two confounders. Second, the event trade itself can affect the volumes and profits of f through channels other than bilateral information sharing. It may have price impact, deplete liquidity, or trigger inventory rebalancing by d or its counterparties, whose rebalancing could involve f . For example, d may partially offload its event trade to another dealer, prompting that dealer to offload its newly acquired inventory to f . Third, common shocks, such as public news or fluctuations in currency demand, can simultaneously affect d 's trade size and f 's trade volumes and profits.

We partial out these two confounders using the unrelated funds as controls. Dealer d would not share private information with an unrelated fund. At the same time, the unrelated funds are, like f , exposed to the price and liquidity impacts of the event trade, heightened trading as the dealer and its counterparties rebalance inventory, and the common shocks. Taken together, comparing f to the unrelated funds preserves all variation in volumes and profits due to information sharing between d and f , while it partials out the variation due to the event trade itself and to shocks common to all funds.

Last, d and f could share data sources and information systems, because they belong to the same banking conglomerate. Our design is meant to detect the increases in f 's volumes and profits that arise through shared databases or information systems, because such sharing is itself a China Wall violation. For example, shared databases are the basis of the SEC's case against Virtu Americas ([US Securities and Exchange Commission, 2024b](#)).

Falsification test. We would falsely conclude that the China Walls are effectively enforced if our design fails to reliably detect MPI sharing although it exists. We test our design's reliability by exploiting the lack of China Walls between nonaffiliated dealers and funds and the stylized fact that dealers extensively share information with their clients ([Barbon et al., 2019](#); [Kumar et al., 2020](#); [Chague et al., 2023](#)). Combining those facts, we

expect nonaffiliated dealers and funds connected via trading relationships to consistently share MPI with each other. We falsify our design if we fail to detect information sharing between the connected dealers and funds.

Never-treated controls. Overlapping events by different dealers can lead an unrelated fund in one event to concurrently be an affiliate or a connected fund in another. Any information shared with the control funds in the concurrent event would partial out the treatment effect and introduce a downward bias. More generally, ever-treated controls yield invalid average treatment-on-the-treated (ATT) estimates that can place negative weights on some groups (Roth, Sant’Anna, Bilinski, and Poe, 2023). For each event, we exclude any unrelated fund that is treated in another event within a 21-trading-day window around the event day, to avoid these threats to inference. Such ‘never-treated’ implementations yield ATT estimates that always place positive weights on all groups (Gardner, 2022).

4.2 Identification Tests

We assume that exceptionally large trades reveal especially valuable private information to the dealer. Section 5.3 jointly tests (I) this assumption, (II) the assumption that our design detects MPI sharing when such sharing exists and does not detect it when sharing does not exist, and (III) the assumption that affiliate dealers and funds do not hide MPI sharing among smaller trades.

We rank each dealer’s trades by USD value from largest to smallest, placing the largest 0.1% of the dealer’s trades in the $[0, 0.1]$ -percentile bin. We define placebo events as dealer-days on which the dealer makes a trade in the $[X, X + 0.1]$ -percentile bin, where $X \in \{10, 20, \dots, 90, 99.9\}$. Our test has two parts. First, we compute the ability of event trades and of placebo event trades to predict price movements in the opposite direction of the dealer, testing the assumption that the dealer’s counterparty is the informed trader. Second, we compare the volumes and the P&L of connected and unrelated funds around each trade-size bin. Under assumption (I), the event trades would predict price movements in the opposite

direction of the dealer, whereas the placebo event trades would not. Under assumption (II), we would detect connected-fund responses around the event trades, and not around the placebo event trades.

The last threat is the possibility that the SEC focuses its enforcement on the largest trades, pushing China Wall violation towards the dealers’ smaller trades. Under assumption (III), we would fail to detect information sharing between affiliate dealers and funds around the placebo event trades.

5 Are the China Walls Effectively Enforced?

[Section 5.1](#) specifies the stacked difference-in-differences regression that implements our design. [Section 5.2](#) presents the results for 2019–2024, the period of tightened China Wall enforcement by the SEC. [Section 5.3](#) tests and rejects two threats to identification. [Appendix C](#) documents heterogeneity in the connected-fund response, consistent with information sharing rather than price or inventory impacts of the event trades.

5.1 Implementation

We adopt the stacked difference-in-differences specification with never-treated controls of [Cengiz et al. \(2019\)](#). Each event creates a stack, a subsample in which an observation is a treated or control fund on a given day from the analytical sample ([Section 3.3](#)) in the 11-trading day window (event window) around the event day. We append the stacks across all events to obtain the sample for regression analysis.

For all regressions, we standardize each dependent variable of fund f by dividing its value by its standard deviation computed over the observations of f . We winsorize these variables at the top 0.5 percent for gross volume, or at both the top and the bottom 0.5 percent for the other variables, after stacking observations and before estimating each regression. All results remain without winsorization.

Our main regression specification is

$$Y_{e(d)ft} = \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^5 \alpha_{\tau} \mathbb{1}_{t=\ell_{e(d)}+\tau} \textit{Affiliate}_{e(d)f} + \delta_{e(d)f} + \varphi_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^5 \gamma_{\tau} \mathbb{1}_{t=\ell_{e(d)}+\tau} + \varepsilon_{e(d)ft}. \quad (1)$$

The dependent variable $Y_{e(d)ft}$ is the gross dollar volume, the net dollar volume in the opposite direction of the dealer's event trade, or the one-week future P&L of fund f (all defined in [Section 3.4](#)) for event $e(d)$ by event dealer d on calendar date t . The affiliate treatment dummy $\textit{Affiliate}_{e(d)f}$ equals 1 if f is affiliated with d . The dummy $\textit{Affiliate}_{e(d)f} = 0$ if f is (i) unrelated to d and (ii) neither an affiliate nor a connected fund in any event within the 21-trading-day window around the event day $\ell_{e(d)}$ ($t = \ell_{e(d)} - 10, \dots, \ell_{e(d)} + 10$). We control for event-by-fund, calendar date, and event-time fixed effects, $\delta_{e(d)f}$, φ_t , and γ_{τ} . We normalize the event-study coefficients by omitting $\tau = -1$ from the event-time indicators and treatment-by-event-time interactions. The event-by-fund fixed effects embed all possible event-by-fund-specific controls, such that our identification comes from differences in within-fund-and-dealer changes around the event day between affiliate funds and unrelated funds. The calendar-date and event-time fixed effects absorb common trends over calendar and event times. We cluster standard errors by event-by-fund and by calendar date, because our treatments are assigned event-by-fund and the incidence of events varies over time. Our clustered variances likely approximate the true variances, since we observe the near universe of trades in the Israeli Shekel, implying a high sampling probability ([Abadie, Athey, Imbens, and Wooldridge, 2023](#)).

The second specification considers the connected funds:

$$Y_{e(d)ft} = \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^5 \beta_{\tau} \mathbb{1}_{t=\ell_{e(d)}+\tau} \textit{Connected}_{e(d)f} + \delta_{e(d)f} + \varphi_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^5 \gamma_{\tau} \mathbb{1}_{t=\ell_{e(d)}+\tau} + \varepsilon_{e(d)ft}. \quad (2)$$

The connected treatment dummy $\textit{Connected}_{e(d)f}$ equals 1 if fund f (a) is connected to d ([Section 3.3](#)) and (b) does not trade with d on the event day or during the five trading days

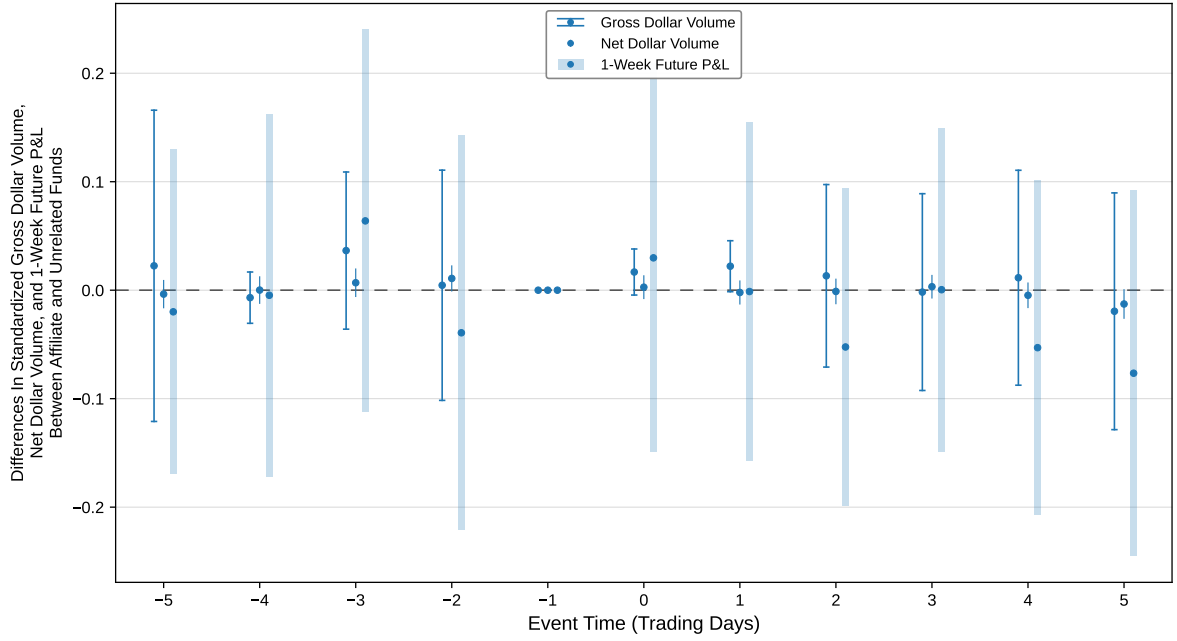
after it ($t = \ell_{e(d)}, \dots, \ell_{e(d)} + 5$). Condition (b) excludes funds whose trades with the event dealer would mechanically change their volumes or P&L around the event day. The dummy $Connected_{e(d)f} = 0$ if f is unrelated to d and is neither an affiliate nor a connected fund in any event within the 21-trading-day window around the event day. All remaining aspects of [equation \(2\)](#) are identical to those in [equation \(1\)](#).

5.2 Results

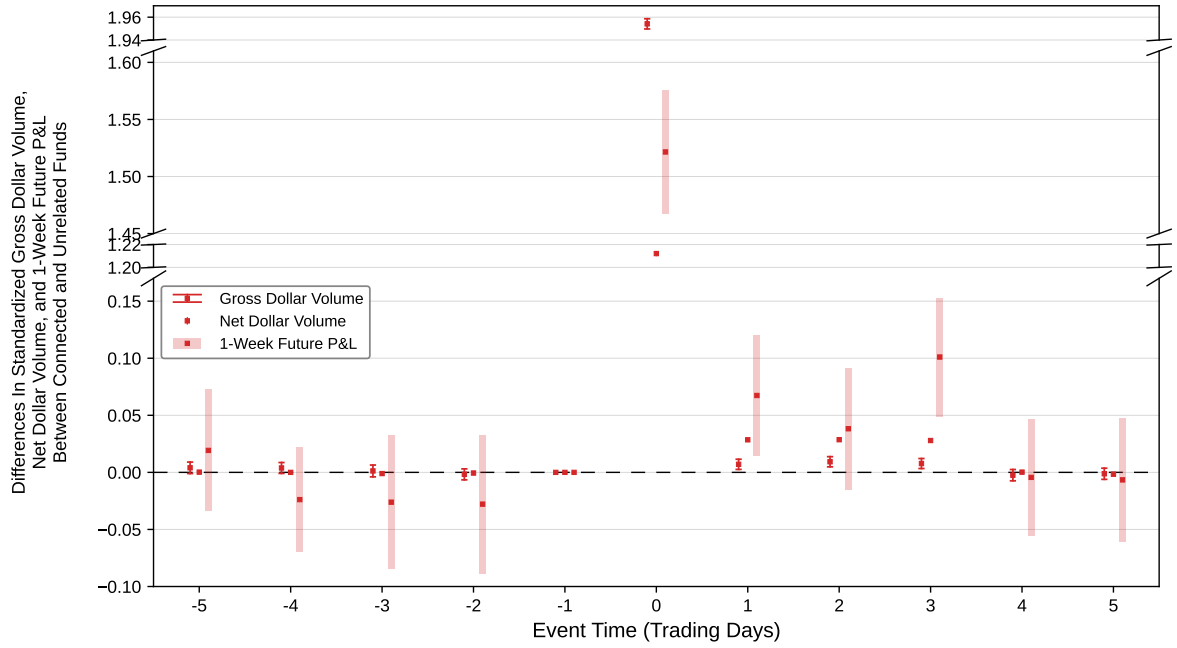
[Figure 5a](#) plots the differences α_τ (in [equation \(1\)](#)) in volumes or profits between affiliate and unrelated funds around the days of exceptionally large trades by dealers in the 2019–2024 period. [Figure 5b](#) plots the corresponding differences β_τ (in [equation \(2\)](#)) between connected and unrelated funds. Each panel displays gross dollar volume (capped confidence intervals), net dollar volume (uncapped), and one-week future P&L (translucent bars). The affiliate funds are indistinguishable from the unrelated funds across all outcomes over the entire event window. The connected funds show no pre-trend, then a large positive response in every outcome on the event day and a weaker positive response for up to a few days thereafter.

More precisely, on each event day, the affiliate funds increase their gross volumes by 0.017 sd (std. error: 0.011 sd) relative to the unrelated funds, and this response remains statistically and economically insignificant on the following days. The connected funds increase theirs by 1.95 sd (std. error: 0.002 sd) on the event day. Likewise, the affiliate funds do not exhibit a detectable increase in their net volumes and P&L on the event day or afterward, whereas the connected funds increase their net volumes by 1.21 sd (std. error: 0.001 sd) and P&L by 1.52 sd (std. error: 0.028 sd), both of which stay elevated for a few days afterwards.

In sum, the affiliate funds are indistinguishable from the unrelated funds around each event trade, while the connected funds trade dramatically more, in the opposite direction of the dealer’s event trade, and earn larger profits on those trades. That the connected funds trade opposite the dealer’s direction is inconsistent with inventory rebalancing driving our results, and that they earn profits on those trades is inconsistent with the dealer being the



(a) Affiliate and Unrelated Funds: Differences in Standardized Volumes and P&L



(b) Connected and Unrelated Funds: Differences in Standardized Volumes and P&L

Figure 5: Coefficient Estimates from (1) and (2) in the 2019–2024 Sample

informed trader on the event trades. We conclude that our design reliably detects the sharing of private order flow information, and that today’s China Walls are effectively enforced.

Table 5: Responses by Funds on and after Event Days

	Affiliate Gross Volume	Affiliate Net Volume	Affiliate P&L	Connected Gross Volume	Connected Net Volume	Connected P&L
$Post \times Affiliate$	-0.01 [0.031]	-0.01 [0.011]	-0.04 [0.040]			
$Post \times Connected$				0.33*** [0.006]	0.22*** [0.004]	0.30*** [0.013]
Event×Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Days-since-Event FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared	0.000	-0.001	0.054	0.081	0.049	0.044
Within R-squared	0.0000	0.0000	0.0000	0.0072	0.0040	0.0010
Observations	116,016,853	116,016,853	115,728,697	124,685,264	124,685,264	124,390,384
Contributing Events	1,318	1,318	1,318	2,632	2,632	2,632
Contributing Dealers	30	30	30	30	30	30
<i>Funds per contributing event:</i>						
Treated (median)	28	28	28	18	18	18
Treated (mean)	51	51	51	325	325	325
Control (median)	3,486	3,486	3,486	3,874	3,874	3,874
Control (mean)	3,523	3,523	3,523	3,820	3,820	3,820

Coefficient estimates from the pooled counterparts to [equations \(1\) and \(2\)](#). *Gross Volume*: Dependent variable is a fund’s standardized daily gross US dollar volume, winsorized at the top 0.5 percentile. *Net Volume*: Dependent variable is a fund’s standardized daily net US dollar volume in the opposite direction of the dealer’s event trade, winsorized at the top and bottom 0.5 percentiles. *P&L*: Dependent variable is a fund’s standardized daily one-week future P&L, winsorized at the top and bottom 0.5 percentiles. An event is a dealer-day on which the dealer makes a trade in the largest 0.1% of its trades by USD value. Each event window is the 11 trading days around the event day. *Affiliate*: Fund is controlled by the same conglomerate as the event dealer. *Connected*: Fund trades at least 10 times with the event dealer in the full 2018–2024 sample, and does not trade with the event dealer on the event day or in the five trading days thereafter. Control funds are unaffiliated with the event dealer, never trade with the event dealer, and are not treated in another event during the 21 trading days around the event day. Contributing events are dealer events with at least one treated fund. Contributing dealers have at least one contributing event. Control observations from events without treated funds also enter the regressions. We include event-by-fund, calendar date, and event-time (days since the event) fixed effects. Standard errors in square brackets are clustered at the event-by-fund and date levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[Table 5](#) details the pooled regression counterpart to [Figure 5](#). It crystallizes our power to detect even tiny treatment effects. Each stacked sample includes many millions of observations corresponding to hundreds of events and numerous affiliate, connected, and unrelated funds per event. Over the 2019–2024 sample, every fund serves as a control or as a connected fund in at least one event. Every dealer-affiliated fund is treated as an affiliate fund in at least one event.

Three of the 46 dealers have no events in the 2019–2024 sample. Another 13 dealers drop out of the affiliate regressions, because they are Israeli or nonbank dealers that do not have affiliate funds. Thirteen dealers drop out of the connected regressions, because nine dealers fall below the BOI reporting threshold ([Footnote 8](#)) and four only trade with nonfinancial firms or with other dealers.

Heterogeneity. [Appendix C](#) examines heterogeneity in the event-day responses in the 2019–2024 sample. The affiliate funds remain indistinguishable from the unrelated funds across every split. The connected-fund responses are strongest for the spot event trades and for the connected hedge funds, consistent with the funds acting on the private order flow information shared by the event dealers. The responses are similar in crisis and non-crisis periods and only modestly stronger for the larger event trades, inconsistent with price or inventory impacts of the event trades driving our results.

5.3 Identification Tests

Three exercises validate our design. We test our assumptions that (I) larger trades reveal more valuable private information to the dealer, (II) our design detects information sharing when such sharing exists and does not detect it when sharing does not exist, and (III) dealers do not hide China Wall violations among their smaller trades.

We first measure how trades of different sizes predict exchange rate returns. We sign each trade from the perspective of the dealer’s counterparty (i.e., opposite to the dealer’s trade direction). Following the intuition that information is reflected in net order flow ([Kyle, 1985](#)), we aggregate signed USD volume within each percentile-rank bin across dealers on each day, obtaining daily *total net dollar volume*. We restrict this exercise to USD/ILS trades to avoid aggregating across currency pairs.

For each trade-size percentile bin $p \in \{[0, 0.1], [10, 10.1], \dots, [90, 90.1], [99.9, 100]\}$ and cumulative return horizon $\ell \in \{0, \dots, 9\}$, we follow a three-step procedure. First, we convert the total net dollar volume on day t into a trade-direction indicator for each bin,

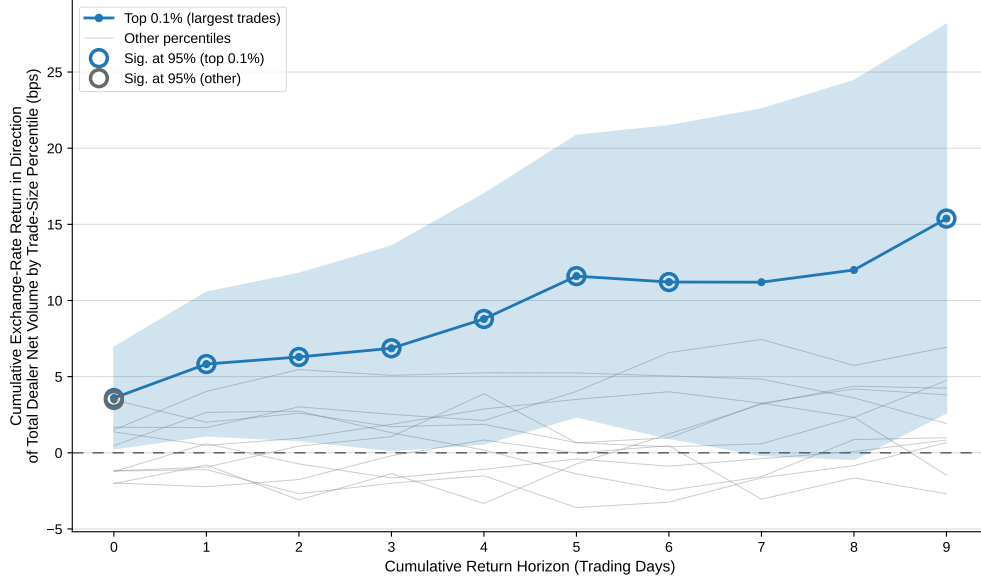


Figure 6: Ability of Dealer Trades to Predict Exchange Rates by Trade-Size Percentile

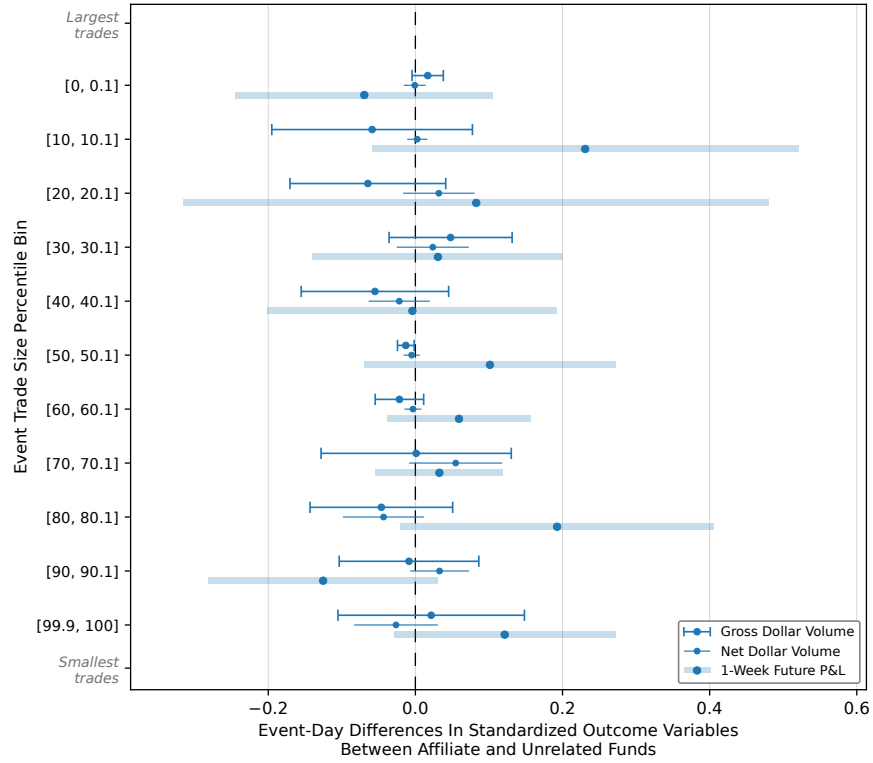
$d_{t,p} := \text{sign}(\text{net volume}_{t,p}) \in \{-1, 0, 1\}$. Second, we calculate the cumulative returns $R_{t-1,t+\ell}$ between $t - 1$ and $t + \ell$ using the Bloomberg USD/ILS exchange rate at 17:00 New York time. Third, the ability of trade-size percentile p to predict exchange rates is computed as the coefficient $\rho_{p,\ell}$ in the time-series regression (3):

$$R_{t-1,t+\ell} = \rho_{p,\ell} \cdot d_{t,p} + \alpha_{p,\ell} + \varepsilon_{t,p,\ell}. \quad (3)$$

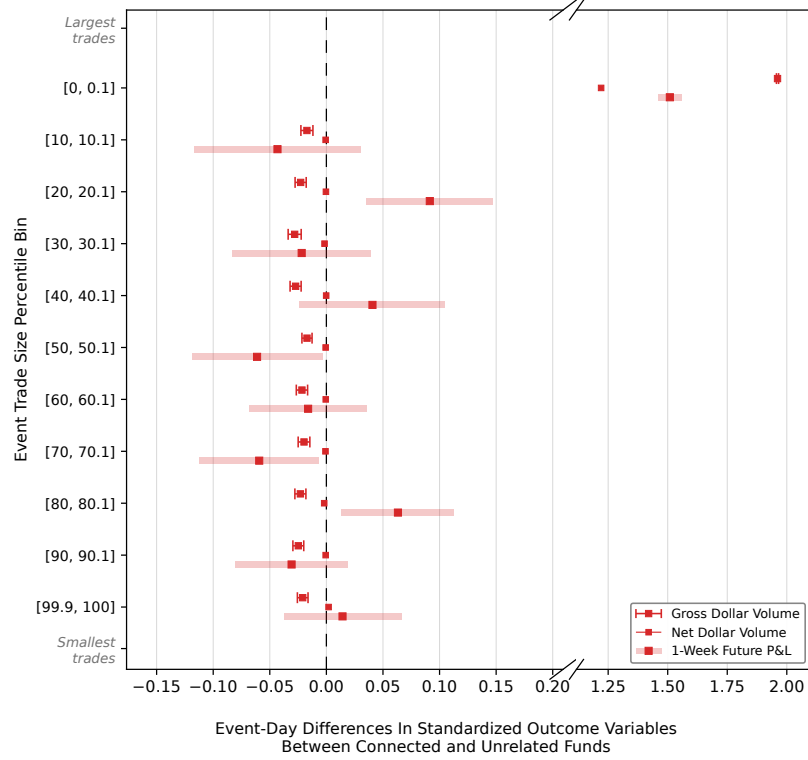
We estimate this regression for each pair (p, ℓ) using OLS with Newey-West standard errors that correct for serial correlation up to ℓ lags.

Figure 6 plots the $\rho_{p,\ell}$ estimates. Consistent with assumption (I), the total net volumes from the event trades (the $[0, 0.1]$ bin) predict cumulative exchange-rate returns up to ten trading days ahead, whereas smaller trades do not. Among all other trade-size bins, only the estimate for $(p, \ell) = ([60, 60.1], 0)$ is significant at the 5% level.

The second and third exercises replicate Figure 5 for each trade-size bin p . The second exercise applies to the connected funds. Figure 7b depicts the event-day coefficient estimates for the connected funds' gross and net volumes and one-week future P&L across p . The



(a) Affiliate-Fund Responses on the Event Day



(b) Connected-Fund Responses on the Event Day

Figure 7: Event-day Coefficient Estimates by Event Trade Size Percentile

estimates are close to zero for every placebo bin except for the event trades already depicted in Figure 5b.¹¹ Combined with Figure 6, and consistent with assumption (II), the connected funds' volumes and profits increase relative to the unrelated funds only around the event trades that are predictive of exchange-rate returns.

The third exercise applies the same specification to the affiliate funds. Figure 7a depicts the corresponding estimates. Consistent with assumption (III), we find no evidence that dealers hide China Wall violations among their smaller trades. The affiliate estimates are economically small and statistically insignificant in every bin, for every dependent variable. In contrast, the identical design detects large and significant responses for the affiliate funds in 2018Q1 and 2018Q2 (Figure 8) and for the connected funds around the event trades (Figure 5b).

6 Did China Walls Reshape Information Sharing?

We investigate the impacts of the SEC's tightened enforcement using the period around the Mizuho settlement on July 23, 2018. We find that dealers cease sharing information with their affiliate funds after the Mizuho settlement, while information sharing between the connected dealers and funds does not change.

Section 6.1 develops the hypotheses and decision rules. Section 6.2 presents the results. Section 6.3 shows that dealers more exposed to US regulation stopped sharing information sooner than other dealers. Section 6.4 tests whether dealers circumvented the China Walls by rerouting information to each other's affiliate funds.

¹¹The connected-fund gross dollar volume estimates hover consistently around -0.02 sd across all placebo percentiles. This pattern arises due to our exclusion of funds that traded with the event dealer on the event day or in the five trading days afterwards. Funds whose gross volumes are increasing around the event day are slightly more likely to trade with the event dealer after the event day, which excludes them more often than those whose gross volumes are declining. This exclusion biases the gross volume estimates downward, working against our main connected fund results.

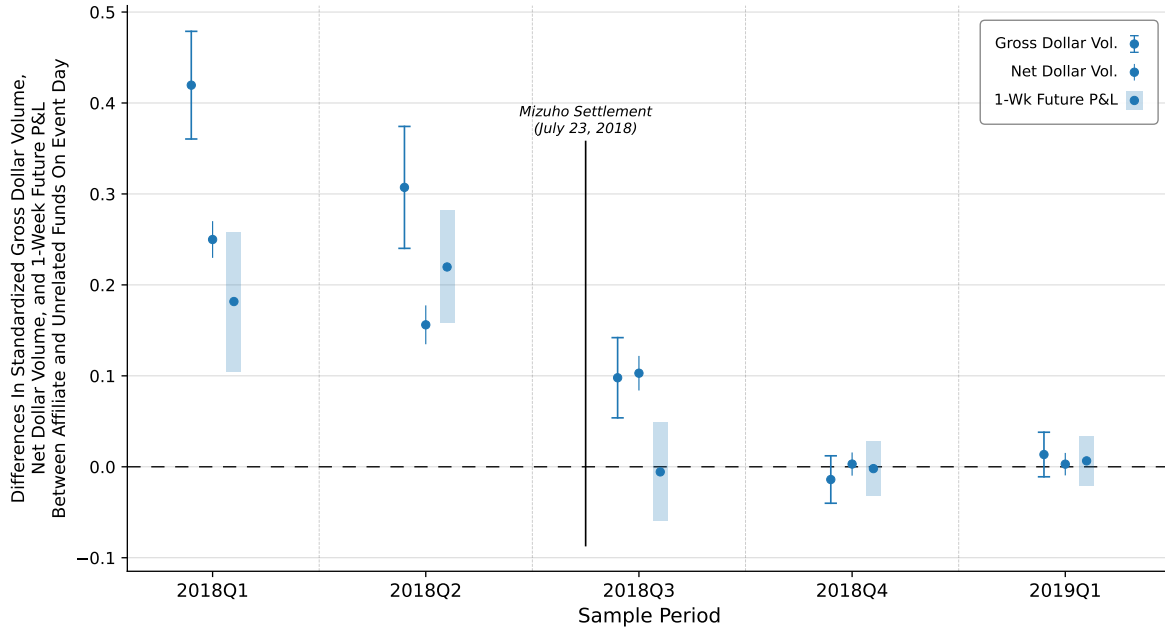
6.1 Hypotheses and Implementation

Hypotheses. Our analysis so far suggests that the SEC is the regulator enforcing the China Walls in our setting, based on its authority over registered broker-dealers (Section 2.1) and on the fact that most ILS trades involve the same dealers as the USD market. We tighten this inference with two tests. First, if SEC enforcement binds, information sharing between affiliate dealers and funds should decline sharply around the Mizuho settlement of July 2018, which signaled tightened enforcement. Second, the decline should come earliest for the US-domiciled dealers, which are most exposed to US regulation.

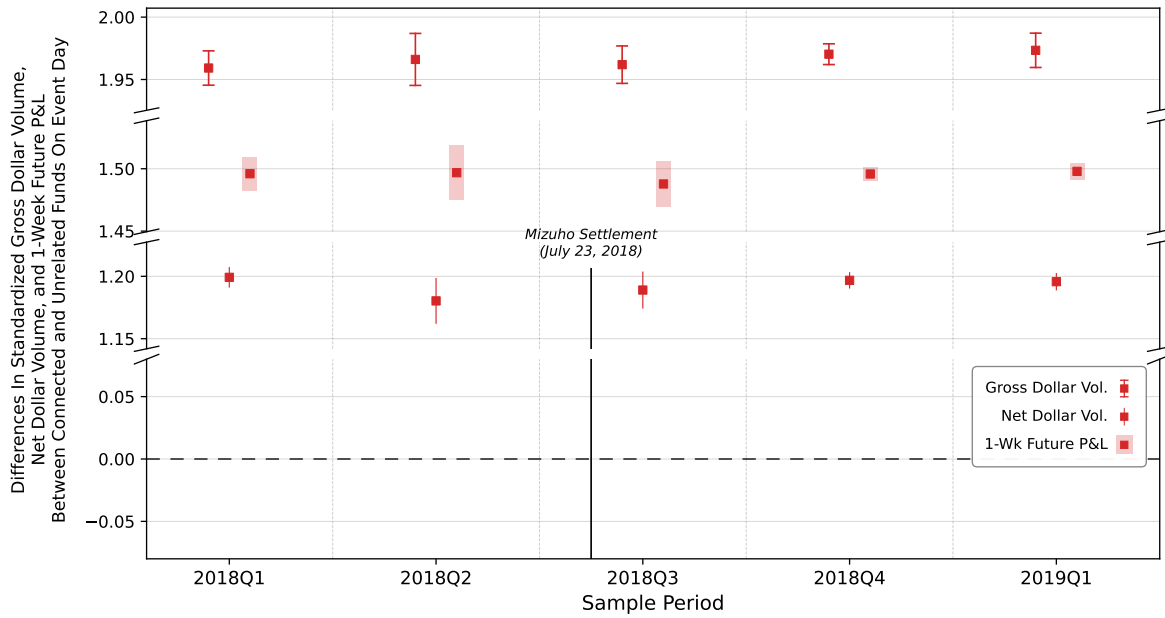
We use the connected funds as a falsification test. The dealers would share less information with their connected funds if the dealers became systematically less willing to share information around July 2018, perhaps due to regulatory changes other than the tightened enforcement of the China Walls. Moreover, the China Walls between affiliate dealers and funds might have spillover effects on how other dealers and funds share information. For example, after the dealers cease sharing MPI with their affiliate funds, other funds trading on the same MPI could earn a larger profit than before. The larger potential profit may induce the dealers to share more MPI with their connected funds. We rule out both broader regulations driving our results and the presence of spillover effects if we fail to detect a change in information sharing between connected dealers and funds around July 2018.

Implementation. We separately estimate equations (1) and (2) for each quarter of 2018 and the first quarter of 2019. We focus on the event-day coefficient estimates, because any response by treated funds concentrates on the event day (as in Figure 5b) and pre-trends are absent (as in Figures 5a and 5b). We assign each event to the quarter containing its event date and retain the full 11-trading-day window, including days that fall outside that quarter. Eight of the 88 events in 2018Q1 fall at the start of the sample and have fewer than five pre-event days. Events, treatments, controls, and the clustering of standard errors follow Section 5.

6.2 Results



(a) Affiliate Funds



(b) Connected Funds

Figure 8: Event-Day Coefficient Estimates from (1) and (2) by Sample Period

Figure 8 plots the resulting event-day coefficient estimates. The affiliate funds increase their gross volumes on the event day by 0.42 sd relative to the unrelated funds in 2018Q1 (std.

error: 0.030 sd) and by 0.31 sd in 2018Q2 (std. error: 0.034 sd). Their net volumes and P&L likewise greatly increase on the event day relative to the unrelated funds. Upon the Mizuho settlement in July 2018, the affiliate coefficients for both volumes and P&L dramatically shrink, then become economically and statistically insignificant from 2018Q4 onwards. The affiliate-fund responses cannot be driven by trading between affiliate dealers and funds, for instance by the affiliate fund taking the other side of the event trade, because we exclude any affiliate fund that trades with the event dealer on the event day or in the following five trading days. The connected funds exhibit large and stable increases in volumes and P&L on the event day across all quarters around July 2018, with gross-volume coefficients between 1.96 and 1.97 sd (Figure 8b), close to the magnitude in the 2019–2024 period (Figure 5b).

A back-of-the-envelope calculation shows that tightly enforced China Walls eliminated \$8.8 million in aggregate affiliate-fund event-day profits, comprising 99.0% of their total event-day profits (\$8.9 million). The dollar value of the eliminated profits is the product of: (a) the average 2018Q1–2018Q2 affiliate coefficient, 0.20 sd, for the event-day difference in one-week future P&L between affiliate and unrelated funds; (b) the average dollar value, \$30,000, of one standard deviation in daily one-week P&L among affiliate funds in the stacked sample used to estimate (1); and (c) the total number, 1,472, of affiliate-fund appearances across event stacks.

6.3 Do US-Domiciled Dealers Respond Sooner to Mizuho?

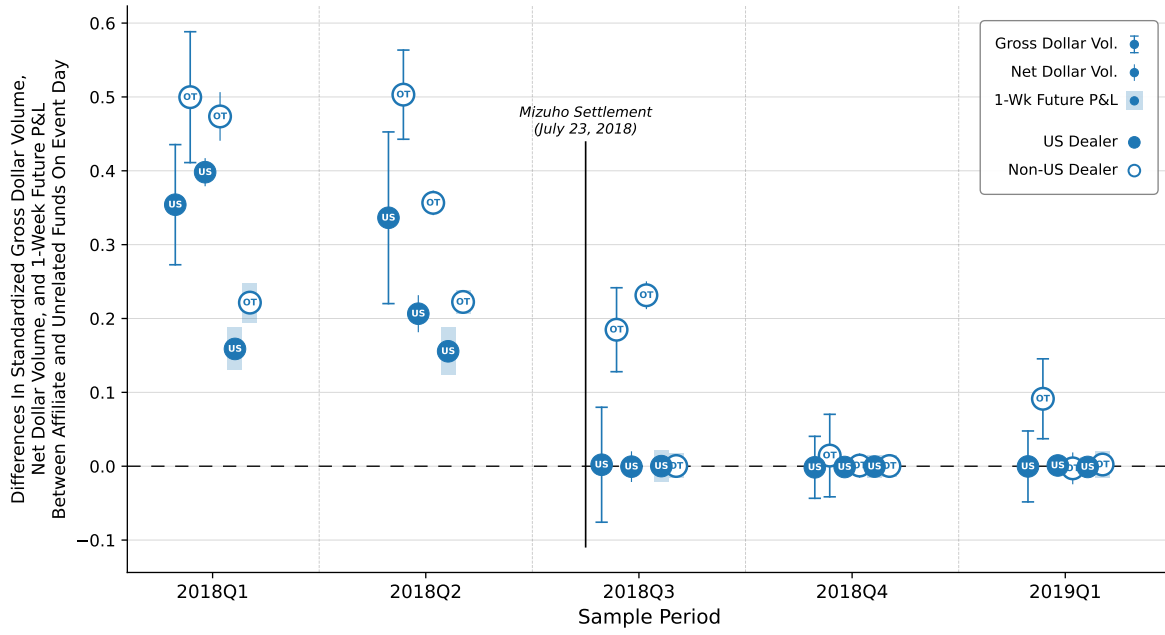
We exploit dealer domiciles to evaluate whether the tightened enforcement around the Mizuho settlement caused affiliate dealers and funds to cease sharing information. The US-domiciled dealers are more exposed to the SEC than the non-US dealers.¹² It follows that, if tightened enforcement by the SEC was the cause, the US dealers would adopt strict China Walls sooner after July 2018 than the non-US dealers.

¹²Armstrong, Glaeser, and Hoopes (2025) use a text-based measure of regulatory exposure within the set of US-incorporated firms. Our approach exploits the majority of dealers in our sample that are not domiciled in the US.

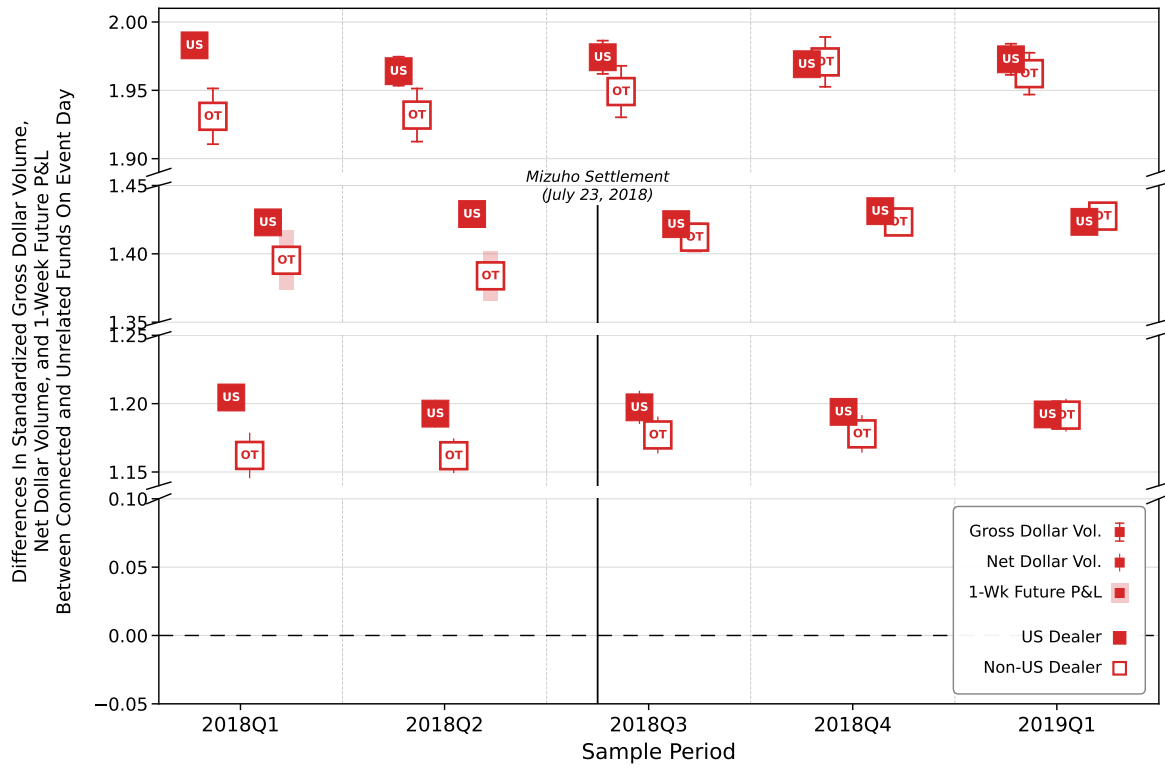
We test this intuition by comparing fund responses to events by the US dealers against those by the non-US dealers around July 2018. We split events into those by US-domiciled and non-US-domiciled dealers, and separately estimate the affiliate regression (1) for each quarter, otherwise replicating Figure 8a. For this split only, we define an event trade as a trade in the largest 0.3% of that dealer’s trades by USD value. We use the 0.3% cutoff rather than the baseline 0.1% cutoff because the quarterly US-dealer subsample would otherwise contain too few events for precise estimation.

Figure 9a plots the estimated coefficients on the event day. Prior to the Mizuho case, affiliate funds respond to events by both US and non-US dealers. They consistently respond less strongly to the US-dealer events, in line with the US dealers’ greater legal liability should the SEC undertake risk-based enforcement. Immediately upon 2018Q3, the affiliate-fund response falls to zero for the US-dealer events, with the gross volume coefficient falling to 0.002 sd in 2018Q3 from 0.34 sd in 2018Q2, and net volume and P&L showing similar patterns. The response to non-US-dealer events remains elevated in 2018Q3 for gross and net volume (0.18 and 0.23 sd, respectively), and becomes insignificant in 2018Q4. The one-quarter delay for non-US dealers is consistent with the tightened enforcement initially binding only for the most SEC-exposed dealers.

Figure 9b splits the connected regression (2) by dealer domicile in the same way. The connected funds’ event-day responses are large and stable across all quarters around the Mizuho settlement, with no discernible change around the settlement for either the US- or the non-US-dealer events. We come to two conclusions. First, the sharp decline in information sharing between dealers and their affiliate funds, coinciding with the Mizuho settlement and arriving sooner for events by US dealers than by non-US dealers, points to the SEC as the regulator enforcing the China Walls in our setting. Second, we rule out both broader regulatory changes driving our results and the presence of spillover effects on connected dealer–fund information sharing, which would each imply a change in information sharing between connected dealers and funds around the settlement.



(a) Affiliate Funds



(b) Connected Funds

Figure 9: Event-Day Coefficient Estimates from (1) and (2) by Dealer Domicile
For this split, an event trade is a trade in the largest 0.3% of the event dealer's trades by USD value.

6.4 Did Dealers Reroute Information to Each Other’s Affiliate Funds?

Favor exchange offers a way for conglomerates to circumvent China Walls. Dealer A shares its private information with Dealer B’s affiliate funds, and Dealer B reciprocates with Dealer A’s affiliate funds. Each dealer’s affiliate funds receive private information from the other dealer while neither dealer breaches its own wall. This circumvention would violate the spirit of the China Walls while technically being in compliance.

Pursuing such favor exchange may increase conglomerate-wide profit. On the other hand, financial firms commonly overcomply with active regulation (Jain, Jain, McNish, and McKenzie, 2013). Favor exchange would imply relatively more information sharing with *conglomerate-affiliated* connected funds (connected funds under a banking conglomerate, each of which owns a dealer) than with *independent* connected funds (those not in a conglomerate, and thus not affiliated with a dealer) after the Mizuho settlement. We would observe rising favor exchange as a widening gap between the responses of conglomerate-affiliated and independent connected funds from 2018 H1 to 2018 H2. A lack of a widening gap would indicate overcompliance.

Figure 10 plots event-day ($t = 0$) coefficients for the connected fund sample, split by whether the connected fund is itself affiliated with another banking conglomerate (C) or is independent (I). The Conglomerate and Independent point estimates are similar in 2018 H1 and 2018 H2 for every dependent variable, with confidence intervals overlapping in every cell. We conclude that overcompliance dominates and that dealers abstain from favor exchange with each other.

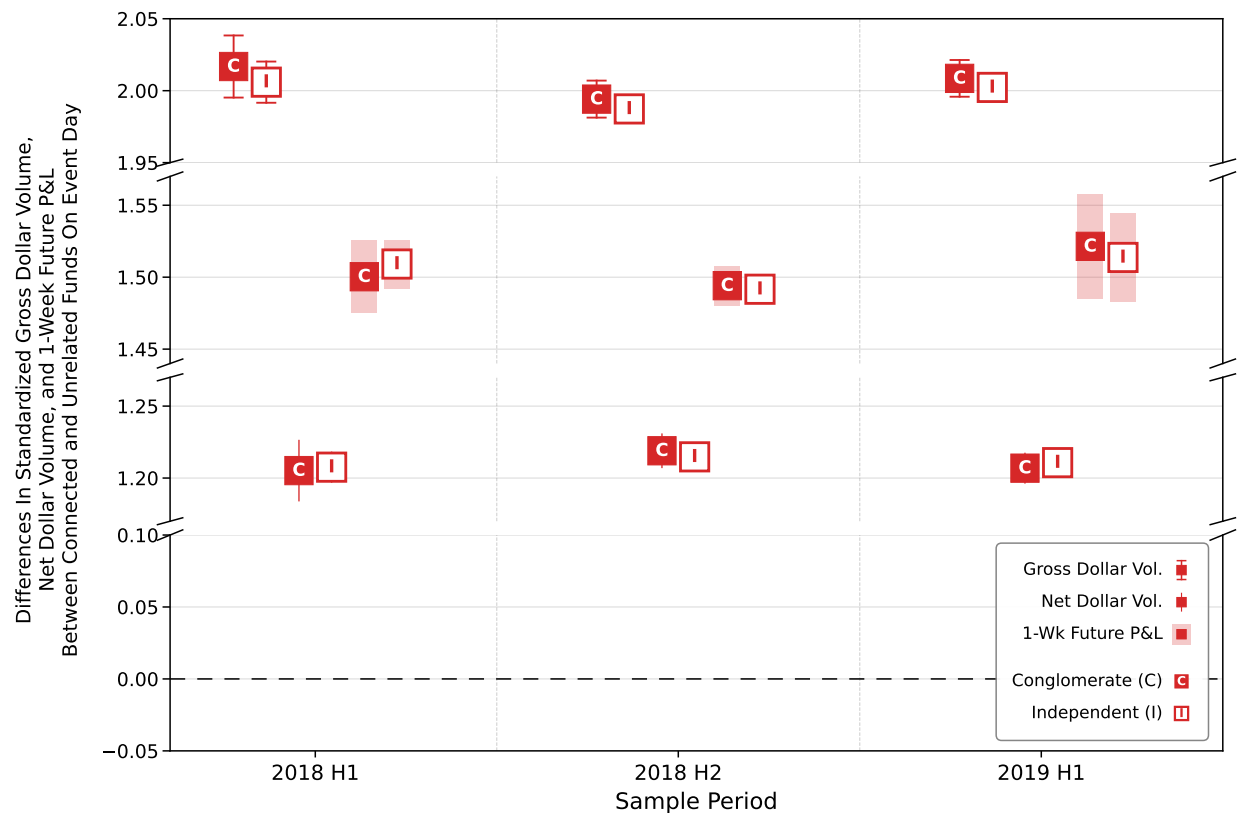


Figure 10: Event-Day Coefficients for Conglomerate-Affiliated vs Independent Connected Funds, by Period

7 How Did Tightened Enforcement Affect Prices and Trading Costs?

We check for three pricing implications one would expect once information sharing between affiliate dealers and funds ceases. First, price discovery would slow, as funds would lose access to their affiliate dealers' private information. Second, conglomerate-affiliated funds would pay lower spreads to their nonaffiliate dealers, as the funds would become less informed and pose a smaller adverse-selection risk. Third, the other funds would pay higher spreads, as dealers less able to monetize the information embedded in client order flow would chase it less aggressively (Pinter, Wang, and Zou, 2025).

Section 7.1 documents suggestive evidence that price discovery slowed after the Mizuho

settlement. [Section 7.2](#) finds that the conglomerate-affiliated funds pay lower spreads after the settlement. [Section 7.3](#) finds that the other funds pay higher spreads after the settlement.

7.1 Did Price Discovery Slow Down?

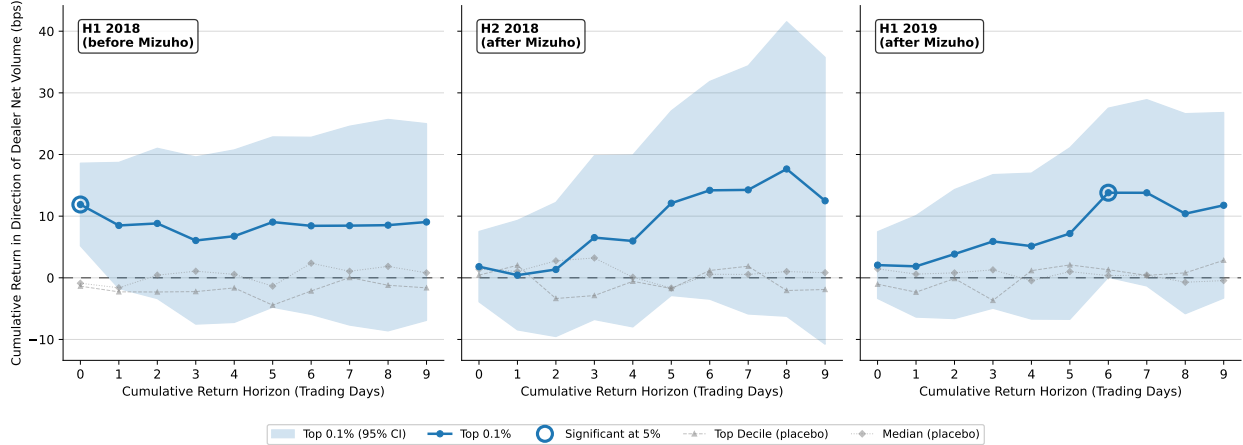


Figure 11: Ability of Dealer Trades to Predict Exchange Rates by Trade-Size Percentile and Period

[Figure 11](#) repeats the price-prediction exercise of [Figure 6](#) separately for each half-year period. It offers suggestive evidence that price discovery slowed after enforcement. In 2018 H1, the cumulative return predicted by the event trades jumps at $\ell = 0$ and stays flat thereafter, consistent with a heavily front-loaded price discovery process. In 2018 H1, only $\ell = 0$ is significant. In 2018 H2 and 2019 H1, the cumulative return builds gradually over the trading week, reaching roughly 11–14 bps at horizons of 6–9 days, tracking the patterns in the 2019–2024 sample ([Figure 6](#)). Across these two periods, only $\ell = 6$ in 2019 H1 is significant.

7.2 Did Trading Costs Fall for Conglomerate-Affiliated Funds?

Funds’ access to private information from their affiliate dealers lets them adversely select other dealers, which would raise spreads in compensation. After the tightly enforced China Walls block this access, the trading costs of the conglomerate-affiliated funds should fall.

We test this hypothesis on a trade-level panel of USD/ILS spot trades. Trades by

conglomerate-affiliated funds are the treatment, and trades by independent funds are the control. The outcome of interest is the effective spread, computed as the difference between the trade price and the dealer’s daily volume-weighted average price, signed from the fund’s perspective. We exclude the trades between affiliate dealers and funds.

We compare the effective spreads of trades by the conglomerate-affiliated funds and by the independent funds before and after the Mizuho settlement.

$$\text{spread}_{i,f(i),d(i),t(i)} = \sum_{\tau \in \mathcal{T}} \alpha_{\tau} \mathbb{1}_{m(i)=\tau} \cdot \text{ConglomerateAffiliated}_{f(i)} + \delta_{f(i),d(i)} + \varphi_{t(i)} + \varepsilon_i. \quad (4)$$

Each USD/ILS spot trade i involves dealer $d(i)$ and fund $f(i)$, and occurs on calendar date $t(i)$ in month $m(i)$. We include $\mathcal{T} = \{-3, -2, 1, 2, 3, 4, 5, 6\}$ months around the Mizuho settlement, with $m = -1$ (June 2018) set as the baseline and $m = 0$ (July 2018) excluded. The dependent variable is the trade’s effective spread, measured in basis points. The dummy $\text{ConglomerateAffiliated}_{f(i)}$ equals one if fund $f(i)$ is affiliated with a banking conglomerate, and equals zero if it is independent. Fund-by-dealer fixed effects, $\delta_{f(i),d(i)}$, and calendar-date fixed effects, $\varphi_{t(i)}$, absorb time-invariant fund–dealer characteristics and shocks common to all trades on the same date. Standard errors are clustered at the fund–dealer level.

Figure 12 plots the monthly α_{τ} from (4) around the Mizuho settlement. Coefficients before July 2018 (April and May 2018) are near zero, supporting parallel trends. Coefficients after July 2018 (August 2018 onward) are negative, with the estimate stable near -4 bps through January 2019.

Table 6 shows that, after July 2018, spreads paid by conglomerate-affiliated funds fell by 4.5 bps relative to those paid by independent funds. Before July 2018, conglomerate-affiliated funds paid 8.2 bps, roughly 127% above the 3.61 bps paid by independent funds in the same window. The reduction is 55% of the conglomerate-affiliated funds’ baseline spread before July 2018. After July 2018, the spreads of the conglomerate-affiliated funds converged

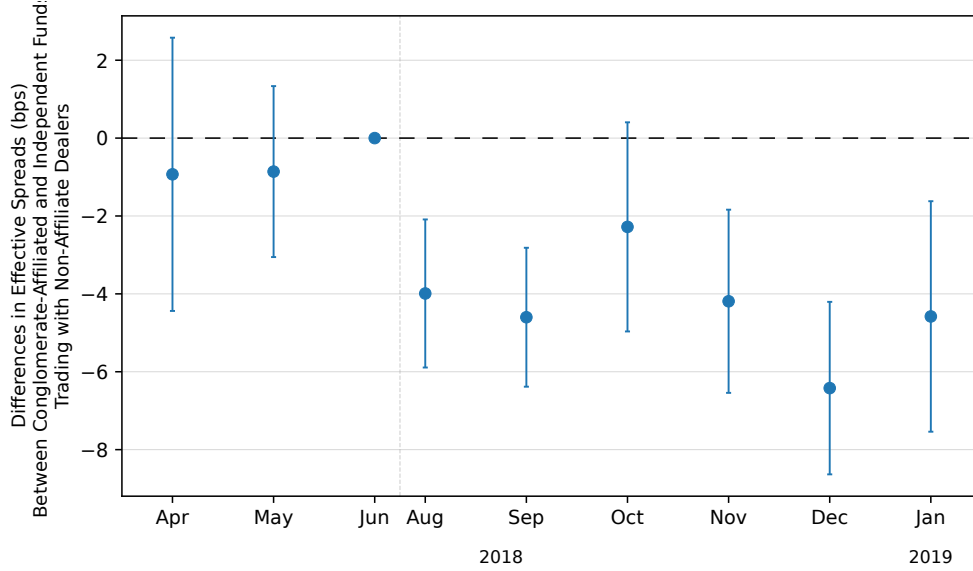


Figure 12: Effective Spreads Paid by Conglomerate-Affiliated Funds Around the Mizuho Settlement

to those of the independent funds, consistent with the conglomerate-affiliated funds losing their informational advantage over other funds.

The implied annual reduction in trading costs is roughly \$1.9M, obtained from the -4.49 bps coefficient applied to the \$4.16B in USD/ILS spot trades between conglomerate-affiliated funds and their nonaffiliate dealers in 2019.

7.3 Did Trading Costs Rise for Other Funds?

We look for spillover effects of tightened enforcement on trades between connected dealer–fund pairs. Effective China Walls prevent a dealer from passing private information gleaned from client order flow to its affiliate funds. The walls therefore cut off one channel through which a conglomerate can convert client information into trading profits. [Pinter et al. \(2025\)](#) predict that dealers chase client order flow to profit from the private information embedded in it. When dealers can no longer monetize this information through affiliate funds, they have less incentive to chase order flow, leading them to raise spreads.

We test our predictions on a trade-level panel of USD/ILS spot trades, as in [Section 7.2](#).

Table 6: Difference-in-Differences Estimates of Effective Spreads for Conglomerate-Affiliated Funds

Treated group	Trades by conglomerate-affiliated funds
Control group	Trades by independent funds
Post $\times$ ConglomerateAffiliated	−4.49*** [0.57]
Pre-period treated mean (bps)	8.2
Pre-period control mean (bps)	3.61
Fund $\times$ Dealer FE	Yes
Calendar Date FE	Yes
Adjusted R-squared	0.051
Within R-squared	0.0008
	All Treated Control
Fund $\times$ dealer pairs	609 44 565
Unique funds	507 41 466
Unique dealers	25 8 25
Observations	43,455 645 42,810

Coefficient estimates are from the pooled counterpart to [equation \(4\)](#). The outcome is the effective spread in basis points, computed as the difference between the trade price and the dealer’s daily volume-weighted average price, signed in the direction of the fund’s trade. The specification compares conglomerate-affiliated funds with independent funds on USD/ILS spot trades, excluding trades between affiliate dealers and funds. The pooled β replaces the dynamic treatment interactions with a single post-period treatment interaction, $\beta \cdot \mathbb{1}_{\text{Post}(i)} \cdot \text{ConglomerateAffiliated}_{f(i)}$. The pre-period is April–June 2018. The post-period is August 2018–January 2019. July 2018 is excluded. The pre-period treated and control mean rows report average effective spreads in the corresponding regression sample over April–June 2018. Every dealer in the sample serves independent funds, and eight also serve conglomerate-affiliated funds, leaving the treated and control unique-dealer counts summing to more than the total. Standard errors in square brackets are clustered at the dealer-fund pair level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We restrict to trades between connected dealers and funds, and compare the effective spreads charged by *conglomerate dealers*, those with an affiliate fund, with the spreads charged by *independent dealers*, those without one.

$$\text{spread}_i = \sum_{\tau \in \mathcal{T}} \alpha_{\tau} \mathbb{1}_{m(i)=\tau} \cdot \text{ConglomerateDealer}_{d(i)} + \delta_{f(i),d(i)} + \varphi_{t(i)} + \varepsilon_i. \quad (5)$$

$\text{ConglomerateDealer}_d$ indicates that dealer d has at least one affiliate fund. The remaining notation is as in [\(4\)](#).

[Figure 13](#) plots the monthly coefficients, which are near zero before the Mizuho set-

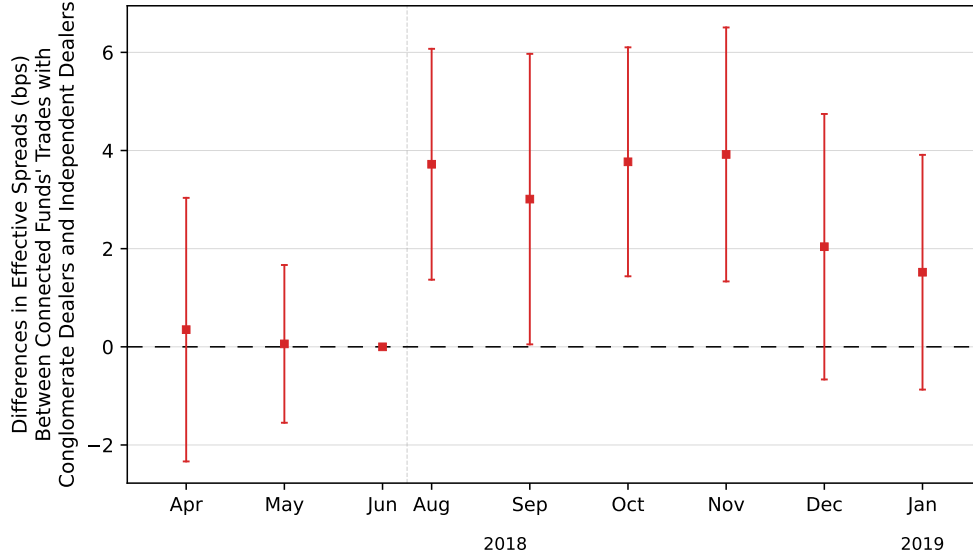


Figure 13: Effective Spreads Charged by Conglomerate Dealers Around the Mizuho Settlement

tlement and positive afterward. Table 7 shows that, before the settlement, conglomerate dealers charged lower spreads than independent dealers. After the settlement, they raised effective spreads by 2.9 bps more than independent dealers. These results are consistent with conglomerate dealers withdrawing the discount they gave their clients, as they chase information less aggressively. While hedge funds are perhaps the most informed clients, we cannot isolate effects on them because every hedge fund in our sample trades only with conglomerate dealers.

Effectively enforced China Walls lower trading costs for conglomerate-affiliated funds and raise the spreads that conglomerate dealers charge. The implied additional annual trading cost is roughly \$46.5M, obtained from scaling the +2.88 bps coefficient by the \$161.5B of gross USD/ILS spot volume that funds traded with their connected conglomerate dealers in 2019. The reduction enjoyed by conglomerate-affiliated funds (Section 7.2) is roughly \$1.9M, much smaller than the increase borne by the other funds. These back-of-the-envelope calculations suggest that effective China Walls raise trading costs in the aggregate.

Table 7: Difference-in-Differences Estimates of Effective Spreads Charged by Conglomerate Dealers

Treated group	Trades with conglomerate dealers
Control group	Trades with independent dealers
Post $\times$ ConglomerateDealer	+2.88*** [0.78]
Pre-period treated mean (bps)	2.51
Pre-period control mean (bps)	3.87
Fund $\times$ Dealer FE	Yes
Calendar Date FE	Yes
Adjusted R-squared	0.065
Within R-squared	0.0000
	All Treated Control
Fund $\times$ dealer pairs	626 561 65
Unique funds	520 474 51
Unique dealers	25 18 7
Observations	45,644 42,817 2,827

Coefficient estimates are from the pooled counterpart to [equation \(5\)](#). The outcome is the effective spread in basis points, computed as the difference between the trade price and the dealer’s daily volume-weighted average price, signed in the direction of the fund’s trade. *Conglomerate dealers* compares trades with conglomerate dealers (those with an affiliate fund) to trades with independent dealers (those without one). The pooled β replaces the dynamic treatment interactions with a single post-period treatment interaction, $\beta \cdot \mathbb{1}_{\text{Post}(i)} \cdot \text{ConglomerateDealer}_{d(i)}$. The pre-period is April–June 2018. The post-period is August 2018–January 2019. July 2018 is excluded. The pre-period treated and control mean rows report average effective spreads in the corresponding regression sample over April–June 2018. Five connected funds trade with both conglomerate and independent dealers, leaving the treated and control unique-fund counts summing to more than the total. Standard errors in square brackets are clustered at the dealer–fund pair level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Conclusion

We document that today’s China Walls effectively block information sharing between affiliate dealers and funds in our foreign exchange setting. Our evidence points to the SEC’s tightened risk-based enforcement around the Mizuho settlement as the cause. Effective China Walls slow price discovery around large dealer trades, lower the spreads paid by conglomerate-affiliated funds, and raise the spreads charged by conglomerate dealers. We conclude that China Walls redistribute informational advantage and reshape competition for client order-flow information.

Our findings reveal a remarkable regulatory capacity to control information flows beyond what prior work has established. Regulators have greatly reduced pollution (Keiser and Shapiro, 2019; Behrer, Glaeser, Ponzetto, and Shleifer, 2021), insider trading (Bhattacharya and Daouk, 2002), misleading financial disclosures (Greenstone, Oyer, and Vissing-Jorgensen, 2006), and discrimination in pay (Bailey, Helgerman, and Stuart, 2024) and in access to accommodation (Cook, Jones, Logan, and Rosé, 2023). In these settings, enforcement occurs to a large extent through private litigation by parties involved in the regulated activity (Glaeser and Shleifer, 2003; La Porta, Lopez-De-Silanes, and Shleifer, 2006). In our setting, a China Wall violation involves affiliates under common corporate control, eliminating the threat of counterparty litigation. Information sharing can also be done in plausibly deniable ways, and bankers often do so (Peluso, 2020). We find that the SEC has the capacity to reshape information sharing even absent the aid of counterparty litigation.

Appendix

A Detailed Context

This section provides detailed institutional context with a focus on the US.

A.1 Definitions

A *banking conglomerate* is a group of firms controlled by the same holding company and that includes a depository institution (i.e., a bank). **Figure 14** summarizes the components of a banking conglomerate. A conglomerate partitions their services into insurers, commercial banks (deposits, loans), investment banks (underwriting, corporate advising), investment funds (asset management), broker-dealers (brokering, dealing, analysis, proprietary trading), and investment advisers.

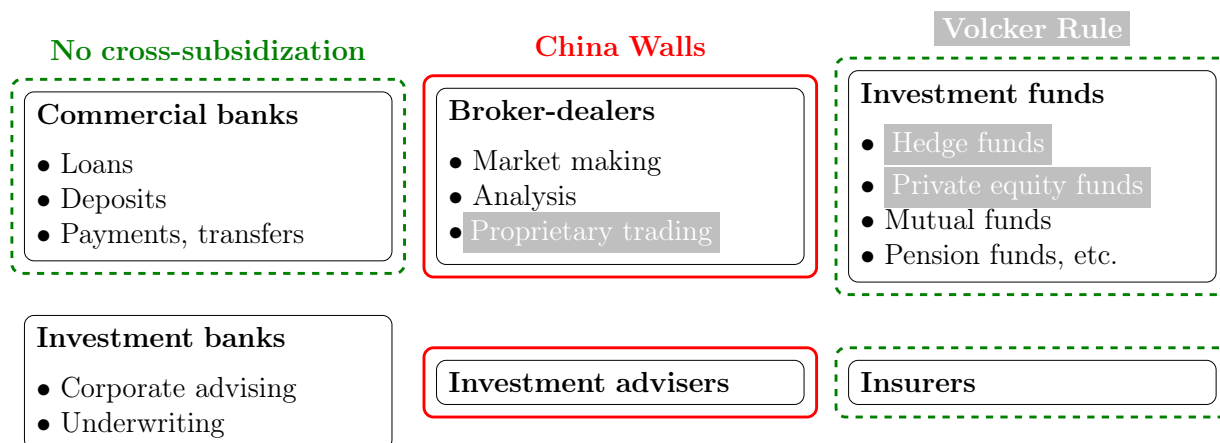


Figure 14: Stylized Banking Conglomerate and Relevant Legal Restrictions

Green dotted lines indicate restrictions on cross-subsidization: banking laws, fiduciary duty to investors, and state-level insurance laws bar commercial banks, investment funds, and insurers from transferring capital to affiliates or trading with them at unfavorable terms. Red solid lines indicate China Walls: broker-dealers and investment advisers are required to prevent their employees from interacting with the employees of affiliates. Black-on-white fonts highlight the Volcker Rule restrictions on proprietary trading and ownership of hedge funds and private equity funds.

All regulations against the misuse or leakage of financial information target material non-public information. MNPI is information that is not publicly disclosed and that reason-

able investors in the relevant securities would find important for their investment decisions. For example, insider earnings information or outstanding order flows of clients are MNPI. Analyses of MNPI are themselves MNPI, whereas private analyses of publicly available information are not. Financial intermediaries owe legal duties over the MNPI of their clients, as we elaborate in [Section A.3](#).

The *China Walls* are blunt internal barriers set around subsidiaries with especially high risk of MNPI misuse. The Walls include both physical barriers and rules, typically:

- Separate offices, elevators, and entry ways for walled-off affiliates, with opaque and soundproof physical barriers when located on the same floor.
- Cool-down periods for employees transferring between walled-off affiliates.
- Watch lists that prohibit employees from trading or advising on the listed securities.
- Records of every instance where an “over-the-wall” executive (who oversees multiple affiliates walled off from each other) receives MNPI from any subsidiary, and requirement that the executive recuse themselves from any business related to the MNPI.
- Monitor and retain all business-related emails and messages sent by employees, and review those containing MNPI.
- Contingency plans when MNPI leaks through the China Walls, and the appointment of officers responsible for enforcing the Walls and handling the contingencies.

These restrictions on employee interactions effectively ban transactions between walled-off affiliates.

A.2 Key Regulations on Banking Conglomerates

The markings in [Figure 14](#) indicate each key regulation on banking conglomerates. Two concerns underlie the regulations. First, the conglomerates may divert publicly insured deposits or insurance premiums towards risky trades or to cross-subsidize affiliates by shifting

risk onto the state or the insureds. Second, the conflicts of interest inherent in combining intermediation, advisory, and trading functions could disadvantage retail investors and undermine trust in financial markets.

Three constraints on banking conglomerates address these concerns. First, a bank or an insurer cannot cross-subsidize affiliates. The US Regulation W (and similar rules elsewhere) limit the outstanding value of bank-to-affiliate transactions to 20 percent of the bank's capital and 10 percent with any single affiliate.¹³ These trades must occur at prevailing market prices and under punitive collateral requirements. Moreover, banks cannot trade securities issued by its affiliates, accept them as collateral, nor guarantee a trade, loan, or securities issuance that involves an affiliate. Analogous rules on insurers, which are harmonized across the US yet enforced by state authorities, prevent their capital being used to subsidize affiliates (Hamilton, 2011).

Second, the Volcker Rule restricts banking conglomerates from proprietary trading and owning risky investment funds. Specifically, a banking conglomerate cannot use its own capital to make short-term profit-seeking trades. The Rule also limits its ownership stake and exposure to hedge funds and private equity funds. Broad exemptions apply. The rule permits any trade held for more than 60 days and trading by broker-dealers that is necessary to support customer-related services (market making, hedging, etc.). Further, hedge funds and private equity funds active entirely outside the US are exempt and, within the US, a conglomerate may sponsor and control such funds if it holds less than 3 percent of the funds' assets. As a result, most banking conglomerates contain hedge funds and considerable scope remains for bank-affiliated broker-dealers to gather private information through market making.

Third, as we elaborate next, the China Walls around broker-dealers and around investment advisers seek to minimize information leakage surrounding these firms. Statutes

¹³Outstanding transaction value include loans, face value of guaranteed assets or liabilities, and gross purchases from affiliates. For example, purchasing \$1 million of an asset from an affiliate would raise the outstanding value by \$1 million until the bank sells \$1 million of the same asset back to that affiliate. (Sales to other affiliates or of other assets do not affect the outstanding value generated by this purchase.)

single out investment advisers for their large potential impact on investment decisions. The broker-dealers are singled out, because their role as intermediaries provide constant stream of privileged information gleaned from their clients' orders. Under the argument that broker-dealers leaking this information to affiliate funds or receiving inside information from affiliates would place the investing public at a sharp disadvantage, preventing such information flows is necessary to maintain trust and participation in financial markets.

A.3 China Wall Enforcement Over Time

Origins. Under common-law tradition, insider trading on behalf of clients was encouraged. Brokers and dealers were expected to use all information that came into their possession, and further solicit inside information, to fulfill their fiduciary duty. This expectation was upended in 1961, when a landmark judgement held each conglomerate liable for damages incurred by the investing public due to trades based on its MNPI. The ruling demands that the intermediaries holding MNPI either publicly disclose or take no action whatsoever related to the MNPI. Subsequent court rulings placed the full burden of avoiding incompatible duties onto the conglomerates.¹⁴

Financial conglomerates were in an impossible legal jeopardy. Beyond fiduciary duty and the new duty to the investing public, the agency principle requires the firms acting as agents to safeguard the private information of their principal (Tuch, 2014). Suppose a conglomerate owns a dealer and a mutual fund, and the dealer receives a large trade request from a client hedge fund—an MNPI. Absent a wall, the law treats the conglomerate as a single entity that knows what each of its parts knows. The mutual fund would then owe its investors the use of this MNPI, under the common-law expectation that intermediaries exploit all

¹⁴A typical case is *Black and Shearson v. Hammill Co.* (Black and Shearson, Hammill Co., 1968) which rules, “conflict in duties is the classic problem encountered by one who serves two masters. It should not be resolved by weighing the conflicting duties; it should be avoided in advance [...] or terminated when it appears.” The judgement upheld awards of \$25 thousand (1968 dollars) each to two customers of a dealer, which sold debentures of a failing firm whose board included a partner at the dealer's parent company. The conflicting duties were the dealer's fiduciary duty to its customers and the partner's duty to keep the inside information of the failing firm confidential.

information in their possession. Yet, doing so would expose the conglomerate to liability if the mutual fund trading on the MNPI cause losses to the investing public. This liability can be avoided only by publicly disclosing the hedge fund’s trade request, in violation of the agency principle. These incompatible duties left financial conglomerates in near-permanent state of legal liability.

In 1968, the SEC offered a way out when it first allowed a conglomerate to settle an enforcement action by adopting a China Wall.¹⁵ The logic is that walled-off subsidiaries can be considered separate entities for the purpose of determining whether a legal duty has been breached. Continuing the example, the dealer would not owe fiduciary duty to the investors of the affiliate mutual fund if the dealer were walled off from this fund. The US financial conglomerates widely adopted the China Walls, which became broadly standardized according to SEC guidelines. Financial conglomerates in other jurisdictions followed, whether through their US operations or regulatory standardization (in Australia, Canada, France, Germany, Japan, Switzerland, and the UK).

Legal history. A 1980 US Supreme Court case (*Chiarella v. United States*) replaced the constellation of duties with one overarching duty to “disclose or abstain.” A person has the duty to disclose or abstain from acting on an MNPI when: (a) she owes fiduciary duty to the source of the MNPI; and (b) the action would give her a personal benefit.

The 1980s also saw the deregulation of financial conglomeration in the US and the UK. The UK removed most restrictions on financial conglomeration in 1986. The US gradually weakened the Glass-Steagall Act provisions throughout the 1980s and 90s, until largely repealing the Act in 1999. The arguments were that full-service financial conglomerates would generate economies of scope and be more competitive against less regulated foreign competitors. Because the duty to disclose or abstain might render full-service conglomerates nonviable, new statutes broadened the legal protection that China Walls provide (Brooke, Burrows, Faber, Harpum, and Silber, 1995, p. 98). Suppose a fund consistently earns large

¹⁵Alternative means to avoid incompatible-duty liabilities, such as obtaining client consent to waive fiduciary duties, are likely ineffective under most circumstances (Tuch, 2014).

profits whenever an affiliate dealer receives large order flows. Under the new statutes, presence of a China Wall between the dealer and the fund would protect the conglomerate against liabilities to the dealer’s clients and to the fund’s counterparties.¹⁶

Pre-1988 regulatory regime. The China Walls were initially a legal benefit available to the banking conglomerates, not a regulatory requirement. As such, the China Walls enforcement was purely reactive, occurring in the course of assigning liability upon the discovery of insider trading. Lacking sufficient China Walls was not an offense in itself. Imposition of large penalties or punishment of individuals required court judgment, with 5-year statute of limitations. A firm that aided a violator could only be prosecuted if the firm knowingly assisted in the violation, a high legal bar. Most importantly, regulatory action required evidence of insider trading. Engaging in practices with a high risk of insider trading was not itself actionable by regulators.

Current Regulatory Regime. Today’s China Walls form a proactively enforced regulatory prescription. The 1988 ITSFEA made the China Walls an affirmative duty for broker-dealers, requiring each to “establish, maintain, and enforce written policies and procedures reasonably designed [...] to prevent the misuse [...] of material, nonpublic information” (§15(g)). The key change was the “risk-based” enforcement powers granted to financial regulators. Rather than requiring actual illegality before the regulators can act, ITSFEA gave them the ability to prosecute behavior that raises the risk of insider trading. Moreover, a regulator can now prescribe corporate organization and internal rules that the regulator believes necessary to cap the risk of illegality to a reasonable level. A regulator can also penalize a firm for an employee’s insider trading when the firm knowingly or recklessly failed to maintain adequate China Walls (§21A of the Exchange Act). A regulator can now levy civil money penalties

¹⁶The China Walls grant similar protection elsewhere. For instance, in a landmark Australian case, *ASIC v. Citigroup (2007)*, Citigroup’s trading arm purchased one million shares of a target firm one day before its acquisition announcement, in a deal where Citigroup’s investment bank was advising the acquirer. The judge dismissed the case, on the basis that the China Wall between Citigroup’s trading and investment bank arms was sufficient to preclude conflict of interest (Hanrahan, 2007).

on registered broker-dealers in its own administrative proceedings, without having to obtain court judgments (the 1990 Securities Enforcement Remedies Act).

B Mizuho Versus Other Cases and Regulatory Changes

We compare the 2018 Mizuho settlement to other China Wall cases across three aspects. *(i)* The regulated parties charged in Mizuho were broker-dealer trading desks, rather than research analysts or investment advisers (Section B.1). *(ii)* The allegations did not involve any trading or profit on the information shared by the trading desks (Section B.2). *(iii)* The legal language was broader than in earlier China Wall cases involving broker-dealer trading desks (Section B.3). We then consider and rule out other salient regulatory changes in 2018 as potential sources of our empirical results (Section B.4).

The Mizuho settlement. In July 2018, the SEC settled with Mizuho over alleged failures to prevent a broker-dealer trading desk from inappropriately sharing client MNPI with another desk (US Securities and Exchange Commission, 2018). From 2012 to 2014, a head trader on Mizuho’s equity trading desk had access to the order-management system of its stock-buyback desk, which handled MNPI about corporate clients’ upcoming buyback orders. The equity desk passed that information to its own customers on several occasions. Mizuho paid \$1.25 million for these failures to maintain its China Walls and to protect client MNPI. The SEC did not allege any trading based on the MNPI, nor any profit to Mizuho from sharing it. In response to this case, Mizuho cut off the equity desk’s access to the order-management system, moved buyback handling to a separate private-side desk on a different floor, and restricted each buyback order information solely to the trader assigned to execute it.

B.1 Regulated Parties and Statutes

China Walls refer to the information barriers required of broker-dealers, of research analysts, or of investment advisers. We study the broker-dealers’ China Walls, which seek

to minimize the risk of insider trading based on their clients’ private order flow, a form of MNPI.

One statute governs the broker-dealers’ China Walls: the Exchange Act §15(g) allows the SEC to prosecute failures to “establish, maintain, and enforce” China Walls around broker-dealers without the need to prove that any insider trading occurred. The Mizuho settlement cites only §15(g). A separate recordkeeping statute, the Exchange Act §17(a), requires broker-dealers to preserve the business messages of their employees. In 2022, Morgan Stanley and Jefferies paid \$125 million and \$50 million, respectively, for failing to preserve the messages their employees sent over personal WhatsApp and Signal accounts ([US Securities and Exchange Commission, 2022](#)). While the recordkeeping duties are legally distinct from the China Walls, broker-dealers routinely use preserved employee messages to enforce related policies.

Table 8: Types of China Walls by Regulated Party and Statute.

Regulated party	Statute	Purpose	Relation to our setting
Broker-dealer	Exchange Act §15(g)	Limit insider-trading risk	our setting
Research analyst	Exchange Act §15D Regulation AC FINRA Rules 2241–2242	Research independence from investment banking	different entity different purpose
Investment adviser	Advisers Act §204A	Limit insider-trading risk	different entity

Table 8 summarizes the legal bases for the China Walls that govern broker-dealers, research analysts, and investment advisers. First, the Exchange Act §15D requires that research analysts be walled off from investment bankers to preserve analyst independence. Each analyst must certify that her reports reflect her own views, and financial firms must identify, manage, and disclose their research conflicts. The 2003 Global Research Analyst Settlement ([US Securities and Exchange Commission, 2003b](#)) found that analysts across ten firms had skewed their research to win investment-banking business for their firms. Those firms agreed

to pay about \$1.4 billion, and their investment bankers were banned from evaluating analysts or setting their pay. The analysts were in turn banned from joining investment pitches and roadshows. Second, the Investment Advisers Act §204A extends to investment advisers the duty to minimize the risk of insider trading. In 2016, Federated Global Investment Management (a registered adviser) was fined \$1.5 million for failing to bring an outside research consultant, who sat on a company’s board, under its insider-trading controls ([US Securities and Exchange Commission, 2016b](#)). These research-analyst and investment-adviser statutes do not apply to the trading desks of broker-dealers.

B.2 Allegations

[Table 9](#) lists, to our knowledge, all broker-dealer trading-desk cases that the SEC or a self-regulatory organization (SRO) charged under the Exchange Act §15(g) (which was numbered §15(f) before the 2010 Dodd-Frank Act), in the order of settlement or final court judgment year. The Mizuho settlement alleges neither trading nor profit based on inappropriately shared MNPI. Nearly every previous case alleges trading, profit, or both. The exceptions are the \$10 million settlement against Morgan Stanley in 2006 and the \$15 million settlement against Citigroup in 2015. Between 1997 and 2006, Morgan Stanley failed to surveil hundreds of thousands of employee and proprietary accounts due to coding errors—251,000 for one such error—and could not verify whether they were trading in watch-listed securities (i.e., those about which it held MNPI). Between 2002 and 2012, Citigroup failed to surveil its trading desks’ securities and swap trades and could not verify whether those desks traded in companies about which it held MNPI, because its data feed only captured loan trades. These surveillance failures were among the longest-running China Wall violations, and no other case involved a greater number of unsurveilled accounts or issuers.

After Mizuho, two broker-dealer trading-desk cases likewise do not allege any trading or profit. In 2021, NYSE Arca, an SRO, settled for \$200,000 with Cantor Fitzgerald, a dealer whose sales and trading desks could access stock-buyback orders that they did not

Table 9: Broker-Dealer Trading-Desk Cases Charged Under Exchange Act §15(g)

Defendant	Year	Trading?	Profit?	Charges beyond §15(g)
Fox-Pitt Kelton ^a	1996	Yes	No	none
J.P. Morgan ^b	1999	Yes	No	fraud (§15(c)(1)–(2), Rule 15c1-2); market-making rules (§11A(c), Rules 11Ac1-1, 15c2-7); recordkeeping (§17(a), Rule 17a-3); supervision (§15(b)(4)(E))
Goldman Sachs ^c	2003	Yes	Yes	fraud (§15(c), Rule 15c1-2)
Blue River Capital ^d	2005	Yes	Yes	fraud (§10(b), Rule 10b-5); recordkeeping (§17(a), Rule 17a-3)
Morgan Stanley ^e	2006	No	No	adviser (§204A)
Friedman Billings Ramsey ^f	2006	Yes	Yes	fraud (§10(b), Rule 10b-5; Securities Act §17(a)); registration (Securities Act §5)
Banc of America Securities ^g	2007	Yes	Yes	fraud (§15(c), Rule 15c1-2)
Merrill Lynch ^h	2011	Yes	No	fraud (§15(c)(1)(A)); recordkeeping (§17(a)); supervision (§15(b)(4)(E))
Citigroup Global Markets ⁱ	2015	No	No	adviser (§206(4), Rule 206(4)-7)
Wolverine Trading ^j	2015	Yes	Yes	adviser (§204A)
Mizuho ^k	2018	No	No	none
Cantor Fitzgerald ^l	2021	No	No	SRO rule (NYSE Arca Rule 11.18)
Morgan Stanley ^m	2024	Yes	Yes	fraud (§10(b), Rule 10b-5(b))
Virtu Americas ⁿ	2025	No	No	none

This table lists, to our knowledge, all cases against broker-dealer trading desks charged under the Exchange Act §15(g) (§15(f) before 2010). *Year*: the year of settlement or final court judgment. *Trading?*: “Yes” if and only if potentially unlawful trading on the MNPI underlying the §15(g) charge was alleged. *Profit?*: “Yes” if and only if illicit profit from that MNPI was alleged. *Charges beyond §15(g)*: all statutes and rules cited in the resolution other than the Exchange Act §15(g) (§15(f) before 2010). ^aFox-Pitt Kelton 1996: several employees traded before the contents of an issuer conference call were made public (US Securities and Exchange Commission, 1996). ^bJ.P. Morgan 1999: market makers manipulated quotes, delayed trade reports, and violated best-execution rules (US Securities and Exchange Commission, 1999). ^cGoldman Sachs 2003: traders bought Treasuries before embargoed refunding information was public (US Securities and Exchange Commission, 2003a). ^dBlue River Capital 2005: its principal served on the bankruptcy committees of issuers whose securities it was trading (US Securities and Exchange Commission, 2005). ^eMorgan Stanley 2006: surveillance failures from 1997 to 2006 left hundreds of thousands of employee accounts unmonitored (US Securities and Exchange Commission, 2006). ^fFriedman Billings Ramsey 2006: market makers shorted shares before an equity offering was announced (US Securities and Exchange Commission, 2007b). ^gBanc of America Securities 2007: traders took positions ahead of forthcoming research (US Securities and Exchange Commission, 2007a). ^hMerrill Lynch 2011: market makers shared client order information with proprietary traders, who traded on it (US Securities and Exchange Commission, 2011). ⁱCitigroup 2015: surveillance failures from 2002 to 2012 left its securities and swap trades unmonitored (a separate adviser charge concerned routing clients’ orders) (US Securities and Exchange Commission, 2015a). ^jWolverine 2015: a market maker shared MNPI with an affiliated adviser, which traded and profited (US Securities and Exchange Commission, 2015b). ^kMizuho 2018: a dealer desk shared client buyback orders with other desks (US Securities and Exchange Commission, 2018). ^lCantor Fitzgerald 2021: traders could access information on stock-buyback orders that they did not handle (NYSE Arca, 2021). ^mMorgan Stanley 2024: clients’ block-trade plans were disclosed to buy-side investors who traded ahead to the benefit of Morgan Stanley, including a profit of \$138 million on the affected trades (US Securities and Exchange Commission, 2024a). ⁿVirtu 2025: proprietary traders could access client-order information through widely shared login credentials (US Securities and Exchange Commission, 2024b, 2025).

handle. In 2025, a consent judgment ordered Virtu Americas to pay \$2.5 million for having a database of customer order information that its proprietary traders could access between 2018 and 2019 ([US Securities and Exchange Commission, 2025](#)). The SEC complaint alleged that Virtu “cannot determine to this day whether its traders abused the trust placed in [it]” by customers ([US Securities and Exchange Commission, 2024b](#), para. 9).

B.3 Legal Language

The SEC has regularly penalized broker-dealers for China Wall violations since the 1990s. Most SEC filings in such cases include a standard statement, quoted here from the 1999 order against J.P. Morgan under the provision’s then-number §15(f) ([US Securities and Exchange Commission, 1999](#)):

The fact that neither the trader [a J.P. Morgan market maker] nor J.P. Morgan is charged with insider trading in violation of the antifraud provisions does not affect the Section 15(f) charge. Section 15(f) by its terms requires reasonable policies and procedures, and an underlying insider trading violation is not a predicate to a Section 15(f) charge.

The Mizuho settlement includes this statement and adds another that broadens “insider” trading to any “unlawful” trading ([US Securities and Exchange Commission, 2018](#), para. 33):

The failure to establish, maintain, or enforce the requisite policies and procedures violates Section 15(g) even if, as in this case, no unlawful trading is alleged to have occurred.

Two earlier orders, which did not concern broker-dealer trading desks, already included a variant of the broadened statement, each writing that these obligations “are important even if no questionable trading has occurred.” First in 2014, a broker subsidiary of IntercontinentalExchange (ICE), the operator of the New York Stock Exchange (NYSE), was charged with failing to wall off its traders from the NYSE’s non-displayed order book ([US Securities](#)

and Exchange Commission, 2014). This charge was ancillary to a broader case against ICE for numerous violations at NYSE of the rules governing stock exchanges. Second in 2016, the research arm of Deutsche Bank was charged with failing to maintain China Walls around analysts (US Securities and Exchange Commission, 2016a). Mizuho was the first trading-desk case to carry the broadened statement.

Mizuho became a reference point in broker-dealer compliance. The 2021 Cantor settlement cites the Mizuho case five times and follows its reading of §15(g) (NYSE Arca, 2021). Moreover, the Securities Industry and Financial Markets Association (SIFMA), the main US broker-dealer trade body, makes Mizuho a leading teaching case in its Compliance and Legal handbook (Barrack et al., 2020). Mizuho is the first case listed in the handbook, ahead of earlier Deutsche Bank, Wells Fargo, Goldman, and Merrill cases.

B.4 Other Regulatory Changes in 2018

Several developments around 2018 affected broker-dealers or the global foreign-exchange market. MiFID II has applied in the European Union since January 2018. This directive governs how EU-authorized investment firms execute transactions in financial instruments and manage their conflicts of interest (European Union, 2026). By May 2018, participants in the wholesale foreign-exchange market were expected to have committed to the FX Global Code, a voluntary code of good practice launched a year earlier (Boge, 2018). Its Principle 19 requires participants to identify and limit access to confidential information, including client order information, its Principle 20 bars disclosing it to external parties absent a valid reason, and its Principle 22 expressly permits dealers to share market color in aggregated and anonymized form (Global Foreign Exchange Committee, 2017). Also in May 2018, the US Congress enacted the Economic Growth, Regulatory Relief, and Consumer Protection Act (EGRRCPA), which relaxed capital supervision for smaller banks and exempted the smallest among them from the Volcker Rule (Congressional Research Service, 2018). Separately, a US federal jury convicted Mark Johnson, the global head of foreign exchange cash

trading at HSBC, of wire fraud in October 2017 for front-running a client’s currency order (U.S. Department of Justice, 2017). Johnson was sentenced in April 2018, in the first criminal conviction of a banker for misusing a client’s order information in the foreign exchange market.¹⁷

None of these developments can account for our results. *(i)* The EGRRCPA loosened rather than tightened regulation. *(ii)* MiFID II excludes spot foreign exchange from its list of regulated assets (European Commission, 2017), whereas our results hold for both spot and non-spot foreign-exchange instruments (Appendix C). *(iii)* The FX Global Code’s Principles 19 and 20 restrict sharing confidential information with affiliates and with clients alike. The Johnson conviction likewise chills sharing with clients as directly as sharing with affiliates. Yet, the decline in information sharing we document is confined to the affiliate dealers and funds, and the connected-fund responses continue unchanged throughout 2018 and onwards. *(iv)* Adherence to the Code was global, and MiFID II is most binding on the EU-authorized firms. The decline in information sharing instead begins with the dealers most exposed to the SEC. Affiliate funds stop responding to events by the US-domiciled dealers one quarter before the non-US dealers (Figure 9a). *(v)* All four developments predate 2018Q3, the quarter of the sharp decline, and the affiliate response remains substantial throughout the first half of 2018, well after the Johnson conviction and the application of MiFID II. In contrast to the alternative candidates, the China Walls separate each dealer’s trading desks from their affiliates, without affecting their ability to share information with their clients.

C Heterogeneity

We examine whether the China Walls hold uniformly across fund and event-trade characteristics in the 2019–2024 sample. Section C.1 describes our implementation. Section C.2 finds evidence consistent with the connected funds acting on the same information that

¹⁷The Second Circuit vacated the conviction in July 2025, after the Supreme Court rejected the relevant theory of wire fraud in *Ciminelli v. United States* (2023).

prompted the event trades. [Section C.3](#) rules out the possibility that price or inventory impacts of event trades, rather than information sharing, are driving the responses of connected funds. Across every split, the affiliate funds remain indistinguishable from the unrelated funds.

C.1 Implementation

We examine fund and event-trade characteristics, and events during crisis and normal periods. Rather than split our sample, we add interaction terms to [equations \(1\) and \(2\)](#) and estimate on the 2019–2024 sample, maximizing our power to detect China Wall violations. We report the event-day ($\tau = 0$) coefficient from this section onward, because any statistically significant specification has the economically largest effects at $\tau = 0$ (as does [Figure 5b](#)).

We define the fund-level dummy $HedgeFund_f = 1$ if and only if the treated or control fund f is a hedge fund. The event-level dummy $SmallEventTrade_{e(d)} = 1$ if and only if the dollar value of the event trade is smaller than the median across all event trades. The event is “Large” if $SmallEventTrade_{e(d)} = 0$. The dummy $Crisis_{e(d)} = 1$ if and only if the event trade occurred during the initial Covid panic, the 2022 Russian invasion of Ukraine, or the 2023 Hamas attack. The dummy $SpotEvent_{e(d)} = 1$ if and only if the event trade (or the largest one if the event has multiple) is a spot trade.

We illustrate the estimation procedure using the large-versus-small event-trade split for the affiliate specification [equation \(1\)](#). First, we add to [equation \(1\)](#) the complete set of interactions between $SmallEventTrade_{e(d)}$ and the event-time and the affiliate treatment dummies, $\mathbb{1}_{t=\ell_{e(d)}+\tau}$ for $\tau = -5, \dots, 5$ and $Affiliate_{e(d)f}$. Second, the event-day ($\tau = 0$) point estimate for the affiliate treatment on the small event trades is the sum of estimated event-day coefficients on terms containing $SmallEventTrade_{e(d)}$ or $SmallEventTrade_{e(d)} \times Affiliate_{e(d)f}$. The large-event-trade estimate is the sum of event-day coefficients on terms containing $Affiliate_{e(d)f}$ that are not interacted with $SmallEventTrade_{e(d)}$. We obtain the

standard errors using the delta method. The other splits work analogously. For example, the hedge-fund split replaces $SmallEventTrade_{e(d)}$ with $HedgeFund_f$.

C.2 Is Information Sharing Driving Our Results?

If the connected funds are acting on private information shared by the event dealer, the connected-fund responses would be stronger (i) for funds with greater capacity to exploit private information, and (ii) when the event trade carries more informational content. We test both predictions using splits grounded in the existing literature on foreign exchange markets.

Hacioglu Hoke et al. (2026) find that hedge funds are more likely to speculate on currency movements and that their trades are more predictive of exchange rates than those of other fund types. If the responses of connected funds reflect information sharing, the hedge funds would respond more strongly than other connected funds. Figure 15, Panel (a), shows that connected hedge funds respond far more strongly than the other connected funds. Nonetheless, hedge funds do not drive the overall connected-fund response, because they are only a small fraction of the connected funds. The other connected funds respond by 1.76 sd in gross dollar volume, comparable to the 2.00 sd average response and highly significant.

Cespa et al. (2022, Table 9) find that spot trades are more predictive of currency movements than forwards and swaps. If the connected funds are responding to the same information as the event dealers, their responses would be strongest for the spot event trades. Panel (a) shows that the responses are indeed strongest for the spot event trades.

These patterns are consistent with connected funds acting on the same information that prompted the event trades. The affiliate funds do not respond even to the events where the connected funds exhibit the strongest responses.

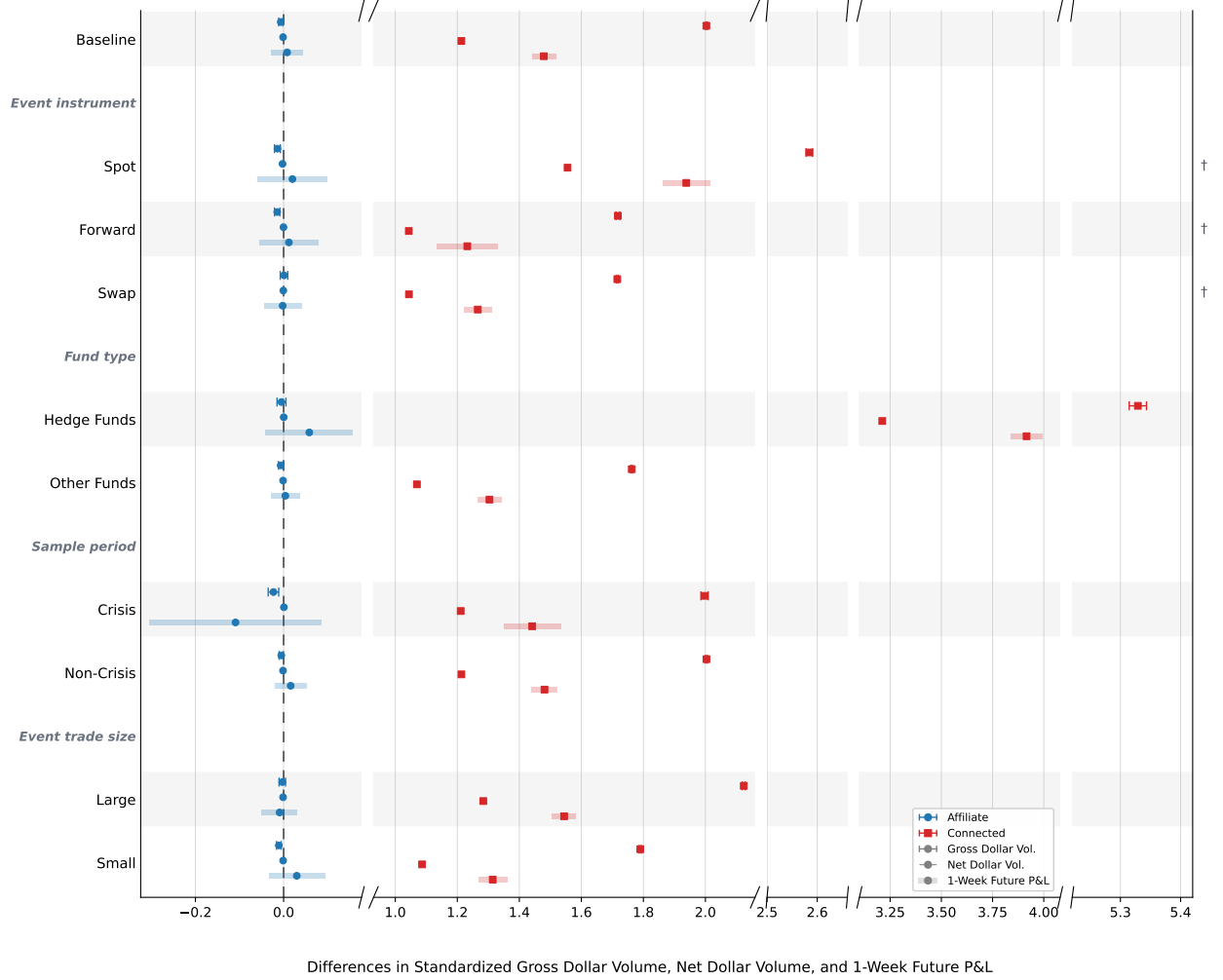


Figure 15: Heterogeneity in Event-Day Fund Responses by Event and Fund Characteristics Panel (a) splits the responses by event instrument and fund type. Panel (b) splits them by sample period and event-trade size.

C.3 Are Price or Inventory Impacts Driving Our Results?

Each event trade may well have price or inventory impacts, which might contaminate the results of our design. If these impacts confound our estimates of connected-fund responses, we cannot conclude that our design reliably detects information sharing. We test in two ways whether these effects survive our unrelated-fund controls and granular fixed effects.

First, the large event trades would have stronger price and inventory impacts than smaller event trades. If those impacts drive the connected-fund responses, we would observe much stronger responses for the large event trades. Figure 15, Panel (b), shows that the connected-

fund responses in gross dollar volume are 2.124 sd for large event trades and 1.790 sd for small ones. The 0.334 difference in responses is consistent with a price or inventory channel, which nonetheless is unlikely to be driving our results.

Second, any impacts on prices or inventories would be amplified when liquidity is thinner, as in crises. More broadly, the top-0.1 percentile cutoff may capture private information of a different type or value during crises, when fund managers shift focus towards market-timing and macroeconomic news (Kacperczyk, Nieuwerburgh, and Veldkamp, 2014; Kacperczyk, Van Nieuwerburgh, and Veldkamp, 2016). Enforcement may also weaken during crises if regulatory resources are more constrained or attention is diverted elsewhere. Crisis-period event trades may also stem from the rebalancing of currency hedges, which moves exchange rates more strongly in volatile periods, rather than from dealer-specific information (Sizova, Syrstad, and Sævareid, 2026). All these possibilities point to a difference in the connected-fund responses during crises compared to non-crisis periods. Panel (b) shows that the connected-fund responses are similar in crisis and non-crisis periods, jointly ruling out all four possibilities.

Taken together, the size and crisis evidence are inconsistent with price or inventory effects driving our results.

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